claude-windows / 6e9d0958-a2ec-4a8e-bb2d-6cce9833c527

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\subagents\agent-a430137c98f9dbf7c.jsonl`
- SHA-256: `df2cbe578b5a7f6b54576a668a6184cc99ffcdd3224c0a1eb7d54f913208a58e`
- Source modified: `2026-07-03T16:41:51+00:00`
- Imported at: `2026-07-05T16:48:26+00:00`
- project: `subagents`
- session_id: `6e9d0958-a2ec-4a8e-bb2d-6cce9833c527`

Transcript

1. user (2026-07-03T16:40:26.331Z)

You are a code-review finder agent. Review target: PR #451 of the badminton-highlight-indexer repo, checked out at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer on branch codex/alpha-readiness-plan (HEAD cd07339). Get the diff with: git diff origin/master..HEAD. Docs-only PR: new docs/ALPHA_LAUNCH_READINESS.md plus index additions.

ANGLE ? altitude. Check the change is implemented at the right depth given the repo's documented doc-lifecycle conventions. Read docs/DOC_STATUS.md (its lifecycle rules and §3a "tracked projects" conventions), docs/README.md structure, and any docs/archives/ conventions. Questions: (1) Does the repo's convention require anything the new tracked doc is missing (e.g., an ADR for a decision like D1-D5 "Decisions locked" ? check docs/archives/decisions/DECISIONS.md for how decisions are normally recorded; is locking new decisions D1-D5 inside a DRAFT tracker doc bypassing the ADR mechanism)? (2) Is a cross-repo tracker placed in one repo's docs/ the right altitude, given the repo set has sibling repos with their own NEXT_STEPS ? does an existing convention say where cross-repo trackers live? (3) The doc's §7 says it is "the source of truth" while NEXT_STEPS.md P0 item remains live ? does this create two competing sources of truth for the same P0 work, and does DOC_STATUS.md convention address that? Flag only concrete convention conflicts you can cite.

Return up to 6 candidate findings as a JSON array of {file, line, summary, failure_scenario}. Return [] if none. Your final message must be only the JSON array.

2. assistant (2026-07-03T16:40:28.457Z)

I'll analyze PR #451 for altitude issues related to the repo's documentation conventions. Let me start by examining the relevant files.

3. user (2026-07-03T16:40:30.006Z)

docs/ALPHA_LAUNCH_READINESS.md
docs/DOC_STATUS.md
docs/NEXT_STEPS.md
docs/README.md

4. assistant (2026-07-03T16:40:31.754Z)

Now let me read the key documentation files to understand the repo's conventions:

5. user (2026-07-03T16:40:32.435Z)

```
1       # Documentation Status & Archival Register
2
3       > **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-diligence**. **Last full review:** 2026-07-02.
4       > **Update cadence:** whenever a doc changes lifecycle state (created / shipped a
```

milestone / delivered / archived), and re-review at each clean-slate.

> **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update it as work finishes** to prove we're done, and (c) **never archive a foundational doc without honestly updating it first** (owner directive, 2026-06-21).

---

0. How to use this
- **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention. §3 is the full register.
- **Marking something complete?** Update the doc's own **§7 progress tracker** (the source of truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2 archival checklist.
- **This register does NOT duplicate completeness detail** ? it indexes lifecycle + archive-readiness and points to each tracked doc's §7.

1. Lifecycle legend
`DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: **LIVING** (permanent reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal record of a run/launch), **CONVENTION** (a standing rule/guide).

2. Archival due-diligence checklist ? *the repeatable process (do NOT just `git mv`)*
Before a foundational/tracked doc moves to `docs/archives/`:
1. **VERIFY claims against the code** ? no stale assertions; tag load-bearing facts `[verified]`. (A doc that says "nothing built" when code shipped is *not* archive-ready ? fix it first.)
2. **Flip Status ? DELIVERED / DONE / REJECTED + a completion note** ? what shipped, where (PRs, paths, artifacts), and what was **deferred / carried forward**.
3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`, `docs/NEXT_STEPS.md`, and any cross-doc links ? the new path + status.
4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the evolution stays traceable.
5. **`git mv` the whole dir/file ? `docs/archives/past_projects/`** (terminal-state ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this in the doc's header).
6. **Record the move** ? flip the row here to §3e + a line in memory `current-state`.

> **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5 delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward WASB-C3/DPA/`md5`), reindexed `docs/README` + `past_projects/README`, kept the ADR-010?014 lineage, then `git mv` to `docs/archives/past_projects/DECOUPLED_COMPUTE/`.

3. The register

3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the completeness source-of-truth)
Doc	Status	Archive-ready?	Next action
`ALPHA_LAUNCH_READINESS.md`	**DRAFT** ? cross-repo alpha readiness tracker created 2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-model/commercial rally-detection gates	No	Execute §7 in order: corpus portability + n=15 baselines, hosted alpha serving, ablation/fusion reruns, then launch GO/NO-GO
`COMPUTE_DECOUPLED_SERVING/`	**IN PROGRESS** ? M0 approved + **M1?M9 LARGELY SHIPPED** (#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0 weights pushed (`weights/0.1.0-l4/`)	No	M11 flywheel + M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap)
`RALLY_QUALITY_RESEARCH.md`	Foundation, **tracked & ACTIVE** (§7)	No	Drives the R-series; complete when §7.1 single quality number is attributed
`R2_EVAL_METRIC_DESIGN.md`	IMPLEMENTED (augmenting)	No (gates R4 + dual-eval; ablation pending)	Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster
`R1_MILESTONE_LAUNCH_REPORT.md`	LAUNCHED (#204) ? terminal record	Soon (with the R-cluster)	Keep with R-series for narrative; archive when the track wraps
`RALLY_DETECTION_GAP_CLOSURE.md`	DRAFT (owner-gated)	No	G1?G3 levers; part of

the R/quality cluster |
39 | `RALLY_DETECTION_FIXES_PLAN.md` | **DONE** ? findings A/B/C resolved; P2b promoted `inactivity_temporal_density` default-ON (#396) | With the R-cluster (deferred per its header; gate = R2 tolF1 ablation + §7.1 number) | Archive with the R-series when the track wraps |
40 | `RALLY_DETECTION_CODE_REVIEW.md` | Terminal snapshot ? findings A (#394), B (#394 ? promoted #396), C (`sdc#50`) all resolved | With the R-cluster | Archive with the R-series |
41 | `PER_VIDEO_WORKSPACE.md` | DRAFT, IN PROGRESS (P0 = #264) | No | ? supersede-reconcile (finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates with M8's v6 cost-ledger ? version bump authored once) |
42 | `MULTI_SIGNAL_FUSION_PLAN.md` | P1 DONE; P2+ owner-gated | No | P3/P4 (learned fusion, audio) ? needs GPU golden features |
43 | `EXPERIMENT_HARNESS.md` | P1 IMPLEMENTED | No | A/B + promotion gate; supports the quality track |
44 | `PROMINENT_COURT_DETECTION.md` | DRAFT (owner-gated) | No | C-series; multi-track selection design |
45 | `EXECUTION_POLICY.md` | P1 SHIPPED; P3b next | No | Async job policy; reused by Cloud Run dispatch |
46 | `GOLDEN_REGRESSION_FIXTURES.md` | IN PROGRESS (owner-gated) | No | Golden eval fixtures |
47 | **Data layer** ? `docs/data_pipeline/` (`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS` · `VIDEO_RECORDING_GUIDELINES` · `GOLDEN_DATA_SHARING`) | LIVING (data, **no-ML**) | n/a | Relocated 2026-06-21 to the isolated data pillar; tracked via `data_pipeline/README.md`. Roora *vendor* docs archived ? `archives/roora-vendor/`. |
48 | `archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md` **(umbrella)** | **? COMPLETE / ARCHIVED (2026-07-02)** ? the five-repo code audit + folded-in *Remediation tracker* (P0?P28 all closed) + absorbed prior audits. All non-owner-gated residuals landed; owner-gated/future items filed as issues (#443/#444/#445, + #301/#327/#332/#384). | Archived | Owner confirmed archival 2026-07-02; moved to `archives/past_projects/` per §2. Residuals now live as GitHub issues, not in this doc. |
49 | `DECISION_SNAPSHOT_REGRESSION.md` | **DRAFT ? decisions LOCKED** (D1?D11 locked 2026-06-25, #383). Zero code shipped for P1?P7. | No | Start P1+P2 (schema + stitch-plan factor); de-risk the P10 main.py split |
50 | `DESKTOP_APP_PLAN.md` | DRAFT (owner-gated, nothing built) | No | P0 compute-feasibility spike; relates to truly-local profile |
51 | `OWNED_MODEL_IMPLEMENTATION_PLAN.md` | PLAN/roadmap (owner-gated) | No (authoritative roadmap) | Gen-0?Gen-2; the owned-model track |
52 | `PERSONALIZATION_PLAN.md` | DESIGN ? owner-adopted; Phase-0 | No | Per-user models (DB v5 store landed) |
53 | `UI_ALPHA_EXECUTION_PLAN.md` | ACTIVE | No | khelsutra UI alpha track |
54 | `I18N_PLAN.md` | Design (P0); **in implementation** (adapted) | No | Executed in `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-gated |
55 | `POST_166_...RISK_ASSESSMENT.md` | Plan for owner review | No | Milestones/risk doc |
56
57 ### 3b. Foundational / strategy references (LIVING ? archive only if **superseded**, not when "done")
58 | Doc | Status | Note |
59 |---|---|---|
60 | `PLATFORM_ARCHITECTURE.md` | Strategy/research | The `ComputeBackend`/`StorageBackend` registries ? realized by COMPUTE_DECOUPLED_SERVING |
61 | `COMMERCIALIZATION.md` | ? LIVING | Strategy; carries WASB-C3 |
62 | `VIDEO_LOCALITY_MODEL.md` | THINKING DOC | L0 fully-local = the truly-local profile |
63 | `HOSTING_PLAN.md` | PROPOSAL | **? partly overtaken by ADR-013 + the shipped site ? verify (finding #5)** |
64 | `UI_PROPOSAL.md` | PROPOSAL (partly shipped) | UX half **in implementation** ? `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc) |
65 | `STORAGE_SHARING_MODEL.md` | DRAFT (co-design pending) | ? |
66 | `ANALYZERS_AND_RECONCILERS.md` | DESIGN (owner-gated) | ? |
67 | `ROADMAP.md` / `CODE_MAP.md` / `HOW_RALLY_DETECTION_WORKS.md` / `KHELSUTRA_PRODUCT_OVERVIEW.md` | LIVING references | Keep current; not archive targets |
68
69 ### 3c. Reports & terminal records (archive candidates once their track wraps)
70 | Doc | Status | Note |

`REAL_FOOTAGE_VALIDATION_REPORT.md`	validation complete	Motivates R2; archive with the R-cluster
`RALLY_DETECTION_QUALITY_REPORT.md`	report	Quality snapshot; part of R-cluster
`QUALITY_ITERATIONS.md`	LIVING empirical log	**Keep** ? the paper-backbone ablation log ([[paper-coauthor-aim]])
`CODE_AUDIT_AND_TEST_HARDENING` · `HARDENING_LOOP_HANDOFF` · `DEFERRED_HARDENING_PLAN`	COMPLETED ? **finish-archived 2026-07-02** (top-level stubs removed; absorbed into the umbrella's *Prior audits*)	? Full copies under `docs/archives/*-COMPLETED-2026-06-14.md`; inbound links repointed (finding #4 resolved)

3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)
`PROJECT_DOC_TEMPLATE.md` · `NEXT_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` · `EVALUATION.md` · `ABLATION_LAYER.md` · `SETUP_NEW_MACHINE.md` · **`CREATING_REELS.md`** (? end-user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) · **`CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md`** (? operator runbook ? `gs://` videos ? reels on L4 Cloud Run; verified e2e 2026-06-23) · `TRACKNET_WSL_SETUP.md` · `README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data_pipeline/`; the field-ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)*

3e. Already archived (`docs/archives/`)
`past_projects/DECOUPLED_COMPUTE/` (? 2026-06-21, #270) · `past_projects/low-regret-moves/` · **`field-pmf/`** (field-ops + customer-feedback, 2026-06-21) · **`roora-vendor/`** (annotation-vendor engagement, 2026-06-21) · `PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md` · `CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` · `DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md` · `HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` · `COMMERCIAL_READINESS_REVIEW.md` · `MONETIZATION_AUDIT.md` · `TEST_RUN_SUMMARY.md` · `research/` · `checkpoints/` · `quality-iterations/` · `past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md` (absorbed ? umbrella) · `past_projects/KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md` (folded ? umbrella).

3f. ADR log ? `docs/archives/decisions/DECISIONS.md`
ADR-001 ? ADR-014 (14, all ratified). The compute/serving lineage: **ADR-005 ? 008 ? 009 ? 010 ? 011 ? 012 ? 013 ? 014**. ADR-014 was **RATIFIED 2026-06-21** (owner M0 sign-off for COMPUTE_DECOUPLED_SERVING).

4. Open due-diligence findings / actions (2026-06-21 review)

> **? Doc-restructuring done (2026-06-21, owner-directed):** field-ops/customer-feedback ? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML **data layer** carved into `docs/data_pipeline/`. Inbound + outbound links rewired; three dir READMEs added (the `data_pipeline` one sets the "no segmentation/stitching" boundary).

1. **? RESOLVED (2026-06-21) ? `PER_VIDEO_WORKSPACE.md` supersedes `PER_VIDEO_OUTPUT_LAYOUT.md`.** Folded forward the `video_slug` naming option (workspace §8.5) + the `output/GX010125_run/` manual-prototype precedent; predecessor archived ? `docs/archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE_MAP.md` repointed.
2. **? RESOLVED (2026-06-21) ? `NEXT_STEPS.md` refreshed** with a current 2026-06-21 state + a **cross-track prioritized pending list**; the 2026-06-13 owned-model/quality body is kept (clearly marked) as that track's detail.
3. **R-series / quality cluster is ACTIVE** (`R1`,`R2`,`RALLY_DETECTION_GAP_CLOSURE`,`RALLY_QUALITY_RESEARCH`,`QUALITY_ITERATIONS`,`REAL_FOOTAGE_VALIDATION_REPORT`,`RALLY_DETECTION_QUALITY_REPORT`). **Do NOT archive yet** ? archive as a *cluster* once R2 is promoted-or-rejected as the gate + §7.1's single number is attributed. (Owner already confirmed "leave as is", 2026-06-21.)
4. **? RESOLVED (2026-07-02) ? finish-archived the hardening stubs.** The three top-level pointer-stubs (`CODE_AUDIT_AND_TEST_HARDENING`, `HARDENING_LOOP_HANDOFF`, `DEFERRED_HARDENING_PLAN`) were removed; full archived copies remain under `docs/archives/`, inbound links (README, PLATFORM_ARCHITECTURE, DESKTOP_APP_PLAN) were repointed, and the sweep is summarized in the umbrella's *Prior audits (absorbed)*.
5. **`HOSTING_PLAN.md` / `UI_PROPOSAL.md` are PROPOSAL but partly overtaken** by ADR-013 (DO dropped) + the shipped khelsutra site. **? `UI_PROPOSAL.md` banner added (2026-06-21)** ? its UX half is now tracked in `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; the `I18N_PLAN.md` localization track is likewise in implementation there. **`HOSTING_PLAN.md` banner still pending** (ADR-013 already noted in it).
6. **? RESOLVED (2026-06-21) ? `COMPUTE_DECOUPLED_SERVING` M0 approved + M1?M4 SHIPPED**

(#279/#280/#282/#283); all §8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized (owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17 $100/?10k cap).
95
96 ## 5. Maintenance
97 Update a row when its doc changes lifecycle; run §2 before any archive; re-review §4 at each clean-slate. This register + the §2 checklist are the operational form of the `CLAUDE.md` archive policy and [[working-agreement-tracked-project-docs]].
98

6. user (2026-07-03T16:40:32.668Z)

1 - RALLY_QUALITY_RESEARCH.md ? **? Foundation doc** for closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps, the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an **adversarially-verified, cited external-research synthesis** (TAL/TAS boundary heads, racket-sports rally-state cues, small-data, eval rigor), a **prioritized cheap?durable roadmap with falsifiable success criteria**, and (§7) a **tracked project plan** ? work-item checklist, sequencing, the single quality **number** attributed to the effort, and a **definition of done** (project completes when all tiers land + the number is recorded). The plan that the QUALITY_ITERATIONS.md log executes against; ties together EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS / OWNED_MODEL_IMPLEMENTATION_PLAN.
2 - RALLY_DETECTION_GAP_CLOSURE.md ? **Tracked execution doc (2026-06-18)** for closing the remaining rally-detection accuracy gap **per video, before launch**, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it: boundaries are tight (R2/R3/R4 holds) ? the gap is **detection**: (G1) recall (local & Gemini both miss ~half on hard clips, and *different* rallies ? complementary), (G2) precision (local over-fires), (G3) **rally-END over-extension on the quick-return-after-point** (shuttle still moving ? needs a *player-state* signal). Levers (owner-gated): player-kinematics + shuttle-in-hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally_gate` Gate A wiring, complementary fusion. **Peer** of `RALLY_QUALITY_RESEARCH.md` (execution tracker, not research); §7 is the source of truth.
3 - PROMINENT_COURT_DETECTION.md ? **Design + experiment (2026-06-19, default-OFF, owner-gated flip)** for the **multicourt G2 over-fire**: identify the **prominent (foreground) court** (the one covering the majority of the frame) and keep only rallies whose shuttle lives inside it ? re-defining the shuttle signal `(x,y,visible)` ? `(x,y,visible-AND-inside-prominent-court)` *before* windowing, so a background/adjacent-court shuttle never forms a rally. 2D floor polygon (MVP) ? **pseudo-3D play-volume** (refinement, so high clears aren't clipped). Born from the GX010142 rally-#1 background-court FP; reuses `fusion_features.point_in_polygon` + `FusionConfig.court_polygon`. §8 progress tracker is the source of truth.
4 - RALLY_DETECTION_QUALITY_REPORT.md ? **Technical-paper checkpoint (2026-06-17)** of rally-detection quality: the honest measured state (over-segmentation milestone tolerance-F1 0.091?0.323 at n=11, p=0.002; merge-rate 0.694?0.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY_QUALITY_RESEARCH.md` + `QUALITY_ITERATIONS.md`.
5 - EVALUATION.md ? How we measure and improve quality.
6 - EXPERIMENT_HARNESS.md ? Design (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with **zero production regression** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.
7 - QUALITY_ITERATIONS.md ? **Living log** of rally-detection quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-chronological record; terminal-state snapshots get archived under `archives/quality-iterations/`.
8 - ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a **signal-fusion framework** for rally detection ? producers emit confidence-scored **signals** at three temporal **scopes** (frame detectors / window analyzers / session analyzers), a learned **scorer** weights them, and an auditable **report-first reconciliation**

pass lints the decision against the evidence. The spine: **no special-casing in the decision path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT_HARNESS.md` and `MULTI_SIGNAL_FUSION_PLAN.md` (ADR-007 modular rally layer).
9 - PERSONALIZATION_PLAN.md ? Design (owner-adopted **hybrid**; Phase-0 landing) for the **per-user personalization feature on a generalization-first baseline**: a thin per-user *tuning* layer gated by the honest small-N statistics (a **min-N promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; *"a tool that gets better the more you help it"*). Records the **locked owner decisions** (identity, storage, `min_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).
10
11 - GOLDEN_REGRESSION_FIXTURES.md ? **Design (owner-gated, NOT started):** video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **§7 progress tracker + definition-of-done**. Extends [GOLDEN_DATA_SHARING.md](data_pipeline/GOLDEN_DATA_SHARING.md).
12
13 ## ? Documentation status & governance
14 - [AUDIT_REPORT_khelsutra-guru_2026-07-02 (? COMPLETE / ARCHIVED)](archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md) ? the five-repo `khelsutra-guru` workspace code audit + its folded-in **Remediation tracker** (P0?P28 all closed). **Archived 2026-07-02** once every remediation wave landed and the last residuals were closed or filed as issues (#443 CORS, #444 corpus Phase-2, #445 GCS eval; + #301/#327/#332/#384). Absorbed the prior `CODE_CLEANUP_AUDIT_2026-06-24` + June-2026 hardening sweep (both archived).
15 - DOC_STATUS.md ? **? living Documentation Status & Archival Register**: every foundational/tracked doc's lifecycle + archive-readiness, the **archival due-diligence checklist** (*honestly update ? then archive*), and the open doc-hygiene findings. Review here; update rows as docs complete.
16
17 ## Product / platform plans
18 - ALPHA_LAUNCH_READINESS.md ? **? Active tracked project (`DRAFT`, created 2026-07-03):** cross-repo alpha readiness tracker after the golden corpus reached 15 videos. Separates the hosted product-alpha lane (shareable upload?process?pick?reel) from the owned-model/commercial rally-detection gate (ship target + WASB-C3 still unresolved). §7 is the source of truth for the concrete next PR sequence: corpus portability, n=15 baseline refresh, ablation/fusion reruns, hosted serving, alpha copy/launch review.
19 - [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? **? Active tracked project (2026-06-21, `DRAFT` owner-gated):** the productionization successor to DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend` (what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU, in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI). Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY proof for $0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a [runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). §7 M0?M10 tracker is the source of truth.
20 - [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ? **DELIVERED + ARCHIVED (2026-06-21).** Lifted the rally pipeline onto a **rented cloud-native GPU** doing bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB runner**; **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ? DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving" subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under `docs/archives/past_projects/DECOUPLED_COMPUTE/` (incl. `DEPLOY_REPORT_GCP_2026-06-20.md`).
21 - PACKAGING_PLAN.md ? **? Active tracked project (`IN PROGRESS`, 11 PRs merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ? `pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra

SPA** single-origin, **torch-free** (LIGHT tier), cross-platform, no native component (decision D10/O1). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP_APP_PLAN.md` narrative.** §7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). **? Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.
22   - DESKTOP_APP_PLAN.md ? **? Superseded narrative (by `PACKAGING_PLAN.md`; D10 = self-hosted webserver, NOT a desktop app).** Original scoping doc (2026-06-18, `[design]`): package the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU* enthusiast laptop (WASB is ~3.4× real-time on a laptop 2070S)? §7 tracker = **P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **`LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.
23
24   ## Practical / ongoing
25   - CREATING_REELS.md ? **? End-user guide:** install the LIGHT-tier app ? `indexer --app` ? process a video ? pick rallies ? download a watermarked reel. Task-first usage walkthrough (web UI + the API chain), with a §6 pointer to the cloud compute option. LIGHT tier only (offline `motion` / optional Gemini; no on-device GPU or one-click cloud). Install mechanics ? root [`README.md`](../README.md); packaging roadmap ? [`PACKAGING_PLAN.md`](PACKAGING_PLAN.md).
26   - CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md ? **? Operator runbook:** `gs://` bucket videos ? highlight reels on the approved **L4 Cloud Run** (scale-to-zero, $0 idle). Copy-pasteable: build image ? create the GPU Job ? cloud-serving control plane ? process(cloud)?query?compile?download ? prove the path (`verify_compute_path.py`) ? $0 teardown. **Verified e2e 2026-06-23.** Raw deploy knobs ? [`../deploy/cloudrun/README.md`](../deploy/cloudrun/README.md).
27   - SETUP_NEW_MACHINE.md ? **? The canonical "spin up a box ? ready for training & inference" runbook.** Copy-pasteable, human-followable, **verified end-to-end on DigitalOcean (2026-06-20)** ? but **compute is now GCP-only (ADR-013)**, see [`deploy/gcp/README.md`](../deploy/gcp/README.md): credentials ? provision ? bootstrap ? suite (922 passed) ? secret-free CPU inference (no GPU/keys) ? real Gemini inference ? Gen-0 training, with pointers to the GPU/native-detector + bucket paths. The `deploy/digitalocean/*` scripts are provider-agnostic and reused on GCP.
28   - DATA_IN_GCS.md ? **? The corpus source-of-truth map: where every piece of data lives in `gs://khelsutra-rally-corpus`** (canonical layout · current contents + drift · how `backend.worker` consumes it · the still-local-only **gap** · migration plan). Read this so a session pulls data from the bucket, not from someone's `C:`/`F:`/Drive ? the goal is to remove the local box as a blocker. Pairs with the two runbooks below.
29   -
[`deploy/gcp/nightly_regression/README.md`](../deploy/gcp/nightly_regression/README.md) ? **? Nightly engine-regression OPERATIONS MANUAL (one-stop shop):** enable · run on demand · change any setting (email recipients/frequency/clips/config) · observe · pause/disable/remove. The full WASB-pipeline reel-count+cost+email nightly on an ephemeral GCP GPU VM (design/status: [`NIGHTLY_ENGINE_REGRESSION.md`](NIGHTLY_ENGINE_REGRESSION.md)). The lighter CPU cached-trajectory tripwire is `scripts/nightly_regression.py` (local Windows Scheduled Task, PR #202).
30   - TRACKNET_WSL_SETUP.md
31   - Evaluation harness and calibration drivers (in `backend/eval/`)
32
33   ## Data layer (collection · annotation · golden sets ? **no ML**)
34   - [`data_pipeline/`](data_pipeline/README.md) ? **the isolated data pillar**: how footage is recorded (`VIDEO_RECORDING_GUIDELINES`), golden sets are created + which videos (`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS`), and data is shared/ingested (`GOLDEN_DATA_SHARING`). Deliberately **no segmentation/stitching/model** detail ? that's the ML pillar (top level + `COMPUTE_DECOUPLED_SERVING/`). The **annotation-vendor (Roora)** engagement is archived under `archives/roora-vendor/`.
35
36   ## Conventions & templates

37 - PROJECT_DOC_TEMPLATE.md ? Reusable skeleton for a **tracked project doc**: status header · decisions-locked · progress tracker (?/?/?) · definition-of-done · archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7 and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.
38
39 ## Historical documents
40 Completed plans, old research, ADRs, and checkpoints live in [`archives/`](archives/).
41
42 - **June-2026 test-hardening sweep** ? **COMPLETED** (2026-06-14; T1?T5 + full loop + verification per #162) and **archived**; the former top-level pointer-stubs were removed 2026-07-02 (absorbed into the umbrella's *Prior audits*). Full records: [`archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md`](archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md), [`archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md`](archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md), [`archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md`](archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md).
43 - **CODE_CLEANUP_AUDIT (2026-06-24)** ? the prior deep engine audit (35 findings; Phase 1 complete, Phase 2 largely shipped, remaining = the `main.py` god-module split). **Absorbed into the umbrella** (see its *Prior audits (absorbed)*) and archived ? [`archives/past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md`](archives/past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md).
44 - **KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN** ? the five-repo audit execution board, **folded into the umbrella** as its *Remediation tracker (live)* and archived ? [`archives/past_projects/KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md`](archives/past_projects/KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md).
45 - DECISION_SNAPSHOT_REGRESSION.md ? **? Tracked project (decisions D1?D11 LOCKED 2026-06-25, #383; implementation P1?P7 pending):** a deterministic **quality-neutral proof** ? freeze + compare the pipeline's **decision output** (rally segments + stitch/concat plan) off a cached trajectory, GPU-free, in CI. The behavioral proof that lets a *genuine* (non-AST-identical) refactor be confidently called quality-neutral. **Never hashes the encoded reel** (non-deterministic). Extends the existing characterization snapshot (`test_golden_fixtures.py`) from segmentation to the stitch plan; peer of the M4 quality-floor + GPU `nightly_reelcount.py`. §7 tracker is the source of truth.
46
47 - **Field-ops / customer-feedback** (recruiting flyer · field-visit playbook · PMF) ? [`archives/field-pmf/`](archives/field-pmf/) (archived 2026-06-21; PMF validation still an active *business* thread ? see `NEXT_STEPS.md`).
48 - **Roora annotation-vendor** engagement (labeling brief · vendor bake-off) ? [`archives/roora-vendor/`](archives/roora-vendor/) (archived 2026-06-21).
49 - Superseded: per-video output layout ? [`archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md`](archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md) (? `PER_VIDEO_WORKSPACE.md`).
50
51 See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md` for classification of small tech debt / starter projects.
52

7. user (2026-07-03T16:40:32.871Z)

1 # Next steps (live) ? refreshed 2026-06-21
2
3 > **How to read:** this top section is the **current cross-track state + prioritized pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed **owned-model & rally-quality track** plan (2026-06-13) is **preserved below, still valid** ? it is now *one of several* active tracks. Live blow-by-blow: memory `current-state.md`. Doc lifecycle/hygiene: `docs/DOC_STATUS.md`.
4
5 ## ? Where we are (2026-06-21)
6 **DECOUPLED_COMPUTE delivered on GCP + archived** (#270); **compute = GCP only** (ADR-013, DO dropped). **`COMPUTE_DECOUPLED_SERVING`** subproject spun up (#272, **ADR-014**) ? one pure core + local/cloud profiles; **M0?M9 LARGELY SHIPPED** (ADR-014 ratified 2026-06-21; M1?M4 #279?#283, M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF). **Per-video workspace P0** shipped (#264); **doc-status register + archival due-diligence** in place (#273). The **khelsutra site + per-rally UI** ship in parallel (sibling track). The **rally-quality /

owned-model** track (below) is the core ML work, gated on **growing the golden corpus**.
7
8 > **? LAUNCH DECISION ? NO-GO (owner, 2026-06-30).** The rally-detection product is
not cleared for an external/commercial launch; we stay in **development**. This is the
expected default for the dev phase. Two unblockers gate a future GO: **(1)** set + meet the
ship-gate target (#172) ? currently unset, needs corpus growth (and the R2 tolF1 ablation,
item 8); **(2)** resolve the **WASB shuttle-detector weight provenance** (commercial-clean
licensing). P2b (#396) was a do-no-harm quality increment, **not** a launch gate ? merging dev
PRs ? launching. Re-decide only when both unblockers clear.
9
10 ## ? Prioritized pending ? all tracks
11 **P0 ? highest leverage / unblocks the most**
12 0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few
recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-
only** (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra
`web/` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable
service (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)**
the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ?
poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290,
default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs
our cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full
record: [[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus
review:** `docs/ALPHA_LAUNCH_READINESS.md` §7.
13 0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable
batteries-included app ? `docs/PACKAGING_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip
install` ? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-
home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades
(P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** **(1)** P3 **GPU
validation + screencast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]`; ?
native-WASB 0-rally tuned-config blocker), **(2)** P2b-install (uv/script + truly-local default
config), **(3)** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then
P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` /
`[[next-session-packaging-app]]`.
14 1. ? **`COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014
ratified). M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged
(default-OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, $0). All §8
gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel
ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12
real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a
$100/?10k cap (D17).
15 2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3
matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus
training feed on it.
16 3. **[owner + Claude · GPU] Full-corpus training (n?5)** ? a real `weights/<ver>/` (the
`0.1.0-l4` bundle is n=2 plumbing-only).
17
18 **P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)**
19 3b. **[Claude] Five-repo audit health remediation ? ? COMPLETE (2026-07-02).** All waves
(P0?P28) landed and the last residuals were closed or filed as issues; the umbrella tracker was
archived to `docs/archives/past_projects/AUDIT_REPORT_khelsutra-
guru_2026-07-02-COMPLETED-2026-07-02.md`. Remaining owner-gated/future work lives as GitHub
issues: #443 (CORS default), #444 (corpus Phase-2 upload), #445 (GCS eval plumbing), plus
#301/#327/#332/#384.
20 4. **[Claude] `COMPUTE_DECOUPLED_SERVING` batch ? REMAINING: M11 + M12**
(M5/M6/M8/M9-auth ? merged this session). **M11** = `backend/flywheel.py` capture?store?shadow-
eval?goldenize plumbing reusing `workspace.py::for_video` + `backend/eval/experiment.compare` +
`backend/eval/promotion.py::promote_if_served_safe`, behind `flywheel.*`=OFF + a faked
`consent_attested` (? privacy-sensitive ? give it fresh focus). **M12** =
`backend/storage/gdrive_api.py` `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme
(regex already allows the hyphen; add to `_KNOWN` + `_backend_class` + the StorageConfig
Literal) against an INJECTED fake Drive (mirror `gcs.py` DI); keep the mount-only `gdrive://`
untouched. ? **M9-consent cols = a new v7 migration** (M8 took v6). M9-consent ToS/DPA text =
owner/counsel.
21 5. **[Claude] Per-video workspace P1?P6** (P1 v6 DB migration ? P2 worker ? P3 per-video
routing of `detector/`+`tmp/` *[the `output/chunks_*` de-hardcode base = ? done via #282]* ? P4

? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.
6. ? **Ops-Agent codified (#291)** ? `deploy/gcp/create_gpu_vm.sh` (the one-command VM helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])
7. **[owner] Set the ship-gate target (#172)** ? needs corpus growth.
8. **[owner-side GPU] R-series: promote-or-reject `R2` tolF1 as the gate** (the ablation) ? needs GPU-extracted golden features.

P2 ? later / gated / deferred
9. Rally-quality **P3/P4** (person-fusion LOVO, audio cue) ? needs GPU golden features.
9b. ? **DONE (#396):** the R4 density fix (finding B / P2b in `RALLY_DETECTION_FIXES_PLAN.md`) is **promoted to default-ON**. Golden-set A/B at the **SERVED** windowing showed tolF1 0.353?0.364 (?tolF1 +0.010, no regression), so `inactivity_temporal_density` default was flipped to True. (Landed flag-gated default-OFF in #394.) Fixed the sparse-tracking quirk where one fast sample after a detection gap held the rally tail open.
10. **M0.4 scorer swap** ? TemporalMaxer (per the 2026-06-17 external review).
11. **Multisport** (table-tennis first, then pickleball).
12. Serving **M8/M9** ? per-video cost ledger, edge auth, DPA/consent (within `COMPUTE_DECOUPLED_SERVING`; owner/legal-gated).
13. **WASB-C3** weight provenance (gates *sharing* a trained model) ? legal.
14. **Rename repo off "badminton"** (owner, BACKLOG).
15. Owner housekeeping: retire the shared **DO droplet** + **DO token**; collector **`md5`** export (collector repo).
16. **PMF field validation** (owner-side, business track).

Doc-hygiene (tracked in `docs/DOC_STATUS.md` §4): ? #1 per-video reconcile + ? #2 this refresh; **pending:** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary confirm); confirm the COMPLETED top-level stubs; verify `HOSTING_PLAN`/`UI_PROPOSAL` currency.

Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)

Updated: 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness design docs merged,
owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-first delivery ? see
`docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**; **PR-2
upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload MD5 byte-identity +
compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep #90?#98 landed** ? ruff/mypy/
coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin collapse into
`trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626 re-verified exact).
Authoritative roadmap:
`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-blow:** memory
`current-state.md` / `session-handoff.md`. This file = the prioritized "what to do next," refreshed each session.

? Next-session / GPU-box pickup (2026-06-15)
M0 in-container quick wins are COMPLETE: ship-gate provisional blocker (**#170**), train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16 drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 / bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level, not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing

? Actions minutes / spending limit).

Immediate next step ? the real P3 LOVO measurement (this box has a GPU + `torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden features + `.rallies.csv` labels are not yet here):
1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6"` ? produces `*_golden_features.json` for the 6 golden videos. (GPU + WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow: **[`GOLDEN_DATA_SHARING.md`](data_pipeline/GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir <shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir <shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).

Also queued (owner-gated / pending input):
- Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
- Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent will do it).
- Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config bug.

Leaving (bulky / M2 ? no start without owner go): ActionFormer (M0.4), event-index DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack deprecation, multisport, repo rename, PMF execution.

Where we are (one paragraph)
ADR-008 ratified ("repo split driven by productionization, not isolation"): training lives in-repo (`training/`)
behind a CI-enforced import wall; the featurizer is single-sourced (`backend/features/rally_seq.py`, train==serve);
the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest output/human_lovo_manifest.json --floor
eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0 (LOVO **0.626**, floor passed,
sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned corpus is in the collector**
(262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet reclassification). The **Gemini
battery is done** (see table) ? the moat is quantified.

The quality table that now drives strategy (IoU 0.5 vs human golden labels)

Path	Mahadevpura (clean-ish)	Kushagra (multi-court)
Heuristic (velocity windowing)	0.657	0.157
Gen-0 owned head (LOVO, n=6)	0.744	0.561
Two-stage WASB+Gemini refine	0.83	?
Gemini wrapper (`gemini-flash-latest`)	**0.93**	**0.66**

Reading: clean footage is commoditized (serve it with Gemini for cents). **Multi-court drops the commodity
wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game pickups + dead-time merges,
the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-growth-sized gap, not an
architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG tapes (uploads candidate clips
only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry load-bearing.
Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json).

Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation

in progress)
95 **This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR #112 + session). All implementation is **owner-gated** (per explicit in the plan); each step = ONE PR off `master`.
96
97 **Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs #156/#157/#160/#161/#163 + #165 + final records).** See archived `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every pre/post/review) and `docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` (full rules + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive done; clean slate.
98
99 - **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion layer) ·
100 `docs/EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) ·
101 `docs/ROORA_LABELING_GUIDELINES.md` (v8) · `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model).
102 - **Key finding (no re-investigation needed):** the owner's "held-shuttle counted as a rally" bug is **NOT a missing capability** ? `backend/pipeline/detectors/rally_gate.py` already implements the net-crossing **"Gate A"** (`apply_gate(..., min_crossings=2)`) that kills service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no `min_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter (now via fusion_hybrid) was the single cheapest, highest-confidence quality win.** (P2 FusionSegmenter specifics: registry `fusion_hybrid`, **default-OFF**, seeds from substrate, persists `net_crossings` + optional person features, optional served Gate A/B via config knobs; adversarially reviewed; suite green. See `tests/test_served_gate_a_regression.py` for the honest finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in with knob for safety, and the leaderboard revert note.)
103 - **Current status (2026-06-14):** P1 (person cue extraction to reusable `PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-ups):** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos; eager-person-compute optimisation); then P4 audio via hit_detector.
104 - **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):**
105 1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served neutral on production windowing per litmus ? kept opt-in with knob for safety.)
106 2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant (players_in_court + two_sided + colocation etc. from `fusion_features`) on the 6 golden; add the eager compute optimisation note/landing. Pure + harness (testable here).
107 3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.
108 - **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default OFF**; promote to default **only on measured LOVO lift** through `run_regression.py` (exit 0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.
109 - ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden features + `.rallies.csv` labels, which are not yet present in this checkout.
110 - **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are exhausted; full details + citations in [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) §5.6.** An owner-commissioned 31-reference review of the quality report *validated* the methodology and surfaced these **future** levers (research inputs, not active work, not new claims; verify citations before any submission):
111 1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as *"the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."*
112 2. **Ground R2 in Action-Spotting / Precise-Event-Spotting** (SoccerNet/TenniSet/T-DEED) ? turns R2 into a defensible publishable contribution.
113 3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).
114 4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free max-pool, ~2.8× fewer GMACs, small-N robust) ? see "M0.4 next increment" below.
115 5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator

noise.

6. **Name the eval?serve lesson "pipeline distribution skew"** ? a citable paper "lesson."

Discipline reminder: owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.

Prioritized next steps

1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every lever below feeds on it.
   Priority order for new videos: **(a) multi-court badminton** (the target distribution AND where the ship target lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1 0.909; pickleball labels are corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool** (consent guardrail). Per video: label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ? re-run the Gen-0 harness.
   The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is reachable this month.
2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480 (necessary), **multi-court reference = wrapper 0.66** (the number that makes the owned model worth serving), clean-footage ceiling = 0.93 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for a lightweight temporal head (MIT, minutes on the 2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold calibration (the LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer TemporalMaxer** (parameter-free max-pool, ~2.8× fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review (`RALLY_QUALITY_RESEARCH.md` §5.6) ? better fit for the cheap/local/fast mandate.
4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`); merged-prototype LOVO across badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated by C3 (provenance multiplies per sport).
5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions: "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists: `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner ratified the physical-visit approach; reframed product-feedback-first per owner spec):** `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner fills ?/contact placeholders and recruits) + `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student script, demo-after-Q6 rule, ground rules, Phase-2 record-and-show-back stretch goal) + `docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md` (G/A/R anchors + **pre-registered decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first visit).
6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified working; confirm old-key deletion in AI Studio); **rename repo off "badminton"** (owner action,

152   in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the battery completed.
153
154   ## Pointers / artifacts
155   - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Topology: ADR-008 (`docs/archives/decisions/DECISIONS.md`) ·
156   Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` · Quality history:
157   `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
158   - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact bundle; weights gitignored,
159   manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
160   - Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest `output/human_lovo_manifest.json`
161   (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\` ·
162   labels: `...\Collect\Golden Labelled\`.
163   - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
164   (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a cold/prepaid project burst-throttles
165   with a MISLEADING "credits depleted" 429; ramp gradually.
166   - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next milestone. Committed next
167   PLATFORM slice = async job model (single-tenant).
168

8. assistant (2026-07-03T16:40:35.375Z)

Now let me read the new ALPHA_LAUNCH_READINESS.md file:

9. user (2026-07-03T16:40:35.821Z)

1   # Alpha Launch Readiness Plan
2
3   > **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:** 2026-07-03; **Last updated:** 2026-07-03
4   > **Lifecycle:** `DRAFT -> IN PROGRESS -> DONE -> archived`
5   > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed in `docs/README.md` and `docs/DOC_STATUS.md`.
6   > **Relation to existing docs:** consolidates the launch-relevant pending items from `NEXT_STEPS.md`, `ROADMAP.md`, `data_pipeline/GOLDEN_VIDEOS.md`, `R2_EVAL_METRIC_DESIGN.md`, `COMPUTE_DECOUPLED_SERVING/`, and the sibling `khelsutra` Studio docs.
7   > **Honesty note:** `[verified]` means checked against the local synced repos on 2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not legal, financial, or launch approval.
8
9   ---
10
11   ## 0. TL;DR
12
13   The corpus is now large enough to move from "grow labels first" into alpha-readiness work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden manifest has 15 badminton videos; `[analysis]` the local manifest needed path rebasing before manifest-driven commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion schema; `[analysis]` the n=15 quality baseline is stale and must be intentionally refreshed before it can gate anything.
14
15   The first controlled alpha should be a hosted, authenticated product loop: upload a video, process it, select rallies, compile, and download a reel. The owned-model/commercial rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the WASB checkpoint provenance is resolved.
16
17   ## 1. Problem & goal
18
19   The previous launch blockers were spread across roadmap docs and mixed together three different questions:

- **Can an invited tester use the product without SSH?**
- **Can the 15-video golden corpus now support honest quality gates?**
- **Can we claim or ship an owned/commercial rally detector?**

This doc separates those into executable workstreams. The goal is to reach an invite-only alpha where the product is usable end-to-end, the quality claims are honest and reproducible, and the remaining model/legal gates are explicit rather than accidentally implied by launch activity.

2. Decisions locked

#	Decision	Source / date	Implication
D1	**Product alpha and owned-model launch are separate decisions.**	`NEXT_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03	We can launch a controlled hosted product loop without claiming the owned model is commercially cleared.
D2	**First alpha must be page-driven, not SSH/CLI-driven.**	`NEXT_STEPS.md` P0 UI-driven/shareable item; verified 2026-07-03	Hosted service, edge auth, upload, job polling, compile, and download are launch-critical.
D3	**The 15-video corpus is now a measurement asset, not a free launch pass.**	`data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03	Rebase/portability, feature completeness, baseline refresh, and gate ratification come before quality claims.
D4	**Do not promote served Gate A by default from the current n=15 evidence.**	`backend.eval.serve_contrast` run, 2026-07-03	Proposal-side lift does not survive served-windowing safety; keep it as an opt-in/analysis lever.
D5	**P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens D13.**	sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03	First alpha can use manual/offline corpus promotion; automated flywheel follows after serving and gates are stable.

3. Verified snapshot, 2026-07-03

All five Git checkouts were fast-forwarded before analysis:

Repo	Branch	Synced HEAD
`badminton-highlight-indexer`	`master`	`e4914d1`
`khelsutra`	`master`	`b0f70a5`
`rally-annotator`	`main`	`ef9e088`
`sports-data-collector`	`master`	`bfc014f`
`sports-obsreport`	`main`	`a0c7252`

Corpus/eval facts:

- `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says `output/human_lovo_manifest.json` is the ingested golden corpus at `n = 15`.
- `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest found all 15 trajectories and labels in this workspace.
- `[analysis]` There are 15 `output/*_golden_features.json` files, but only 6 carry full shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped by fusion/person eval.
- `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` loaded only 6 videos and measured person-fusion `Delta F1 = -0.021`; not promotable.
- `[analysis]` `python -m backend.eval.serve_contrast --manifest <rebased-n15>` measured proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.
- `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor eval_baselines/heuristic_lovo_n6.json` produced learned LOVO mean F1 `0.551`, `ship=False`; blockers remain stale n=6 floor/sign test, uncalibrated ship target, and WASB C3 provenance.
- `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15> --features-dir output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off tolF1 `0.1382`, but marked the frozen baseline stale due to config/corpus changes.
- `[analysis]` `python -m backend.eval.ablation --features-dir output --granularity group` currently fails at import due to a circular import between `backend/eval/ablation.py` and `backend/eval/ablation_pdf.py`.

4. Launch strategy

4.1 Alpha lane: make the product usable

The alpha lane is successful when an invited tester can open the app, authenticate, upload a video, watch the job progress, pick rallies, compile a reel, and download it. This lane should avoid model-quality promises beyond the measured state and should default to the most reliable serving path.

4.2 Quality lane: make the 15-video corpus gate-ready

The quality lane is successful when the 15-video corpus is portable, reproducible, feature-complete, and backed by refreshed baselines. Only then should R2 tolF1, Gen-0/TemporalMaxer, or other model/promotion decisions be ratified.

4.3 Flywheel lane: make tester feedback useful

The flywheel lane is successful when user corrections can flow into the collector/vault path with consent/provenance and a visible re-eval delta. This is valuable for alpha learning, but it should not block the first controlled alpha if manual promotion can bridge the gap.

5. Risks and guardrails

Risk	Why it matters	Guardrail
Launch activity implies model launch approval	The engine docs explicitly say rally-detection product launch is NO-GO until ship target and WASB C3 clear	Keep all alpha copy scoped to invite-only processing, not owned-model claims
Stale local manifests create false confidence	Manifest-driven tools skipped or failed until paths were rebased	Make the corpus manifest portable and add a path/fixture smoke check
15-video corpus is only partially feature-complete	Fusion/person/audio eval is still n=6 today	Regenerate or upconvert full-fusion features for all 15 before fusion decisions
Baseline refresh accidentally rubber-stamps a regression	Nightly runner correctly reported stale baselines	Refresh baselines only after reviewer sign-off and record the command/output
P10/P11 expands scope before hosted alpha is usable	Corpus flywheel is valuable but can consume the alpha launch window	Keep P10/P11 behind explicit owner reopen unless alpha lane is already stable
WASB checkpoint provenance remains unresolved	Blocks commercial sharing of a trained model regardless of F1	Treat C3 as a hard model-release gate; do not hide it inside product launch tasks

6. Honest limits

Completing this doc's alpha lane does **not** prove the owned model is ready, that WASB weights are commercially clean, that the product can serve unbounded public traffic, that multi-sport is ready, or that every tester correction automatically improves the model.

Completing the quality lane does **not** by itself deploy a service. It only makes the measurement surface honest enough for a launch decision.

7. Deliverables & progress tracker

Legend: TODO = not started, IN PROGRESS = actively being worked, DONE = shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the row explicitly names a cross-repo batch.

ID	Deliverable	Repo / owner	Depends on	Gated?	Status	PR
A0	Create this alpha readiness tracker and index it in docs	`badminton-highlight-indexer`	-	No	IN PROGRESS	#451
A1	Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check, edge auth ON, upload -> process -> jobs -> compile -> download verified	`khelsutra` +				

`badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |

A2	Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that fails loudly when labels or trajectories are missing	`badminton-highlight-indexer`	A0	No	TODO	-
A3	Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile collector's six-video source manifest with the 15-video eval corpus	`sports-data-collector` + vault	A2	Owner upload/storage scope	TODO	-
A4	Refresh the n=15 nightly regression baseline intentionally and record the exact command/output; keep stale-baseline warnings until reviewed	`badminton-highlight-indexer`	A2	Reviewer sign-off	TODO	-
A5	Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6` floor for future Gen-0 comparisons	`badminton-highlight-indexer`	A2	No	TODO	-
A6	Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the n=15 corpus	`badminton-highlight-indexer`	A2, A4	No	TODO	-
A7	Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so fusion/person/audio analyses run over all 15	`badminton-highlight-indexer`	A2	GPU/data availability	TODO	-
A8	Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly promote/reject person fusion, Gate A served-default, and any other R-series default flips	`badminton-highlight-indexer`	A6, A7	Reviewer sign-off	TODO	-
A9	Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-serving artifact bundle and gate verdict	`badminton-highlight-indexer`	A5, A8	Owner approval for M0.4 scope	TODO	-
A10	Decide and record the alpha ship-gate target: metric, corpus slice, minimum threshold, significance rule, and what claims are allowed	Owner + `badminton-highlight-indexer`	A4, A5, A8	Owner decision	GATED	-
A11	Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred, document "no commercial model sharing" copy guardrail	Owner/legal	A10	Legal/provenance	GATED	-
A12	Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion, `train|reeval` job, and delta-F1 surface	`khelsutra` + `badminton-highlight-indexer` + collector	A1, A3	Owner D13 reopen	GATED	-
A13	Alpha launch review: runbook, privacy/consent copy, known limits, rollback/disable steps, and owner GO/NO-GO	Cross-repo	A1, A4, A10, A11	Owner launch decision	TODO	-

8. Concrete next PR sequence

1. **Docs PR:** land this tracker.
2. **Corpus portability PR:** rebase/portable manifest tooling plus missing-file smoke test.
3. **Hosted alpha PR/batch:** health/auth/upload/process/job/compile/download verification and runbook updates.
4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor refresh.
5. **Ablation/fusion PRs:** fix ablation import, complete 15-video features, rerun promotion/rejection decisions.
6. **Launch decision PR/docs update:** record ship-gate target, C3 status, alpha copy limits, and launch checklist.

9. Open questions

1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or should P10/P11 be reopened before any tester wave?
2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU, GCE L4 VM, or another approved GPU host?
3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains unresolved?
4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a checked-in redacted manifest, or both?
5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every clip, or should the eval loaders support mixed schemas with explicit capability tags?

10. Definition of done

- [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without SSH.
- [ ] Alpha service is gated behind the chosen auth posture and has a documented disable/rollback path.
- [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely on stale local absolute paths.
- [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
- [ ] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
- [ ] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected on the 15-video corpus.
- [ ] Ship-gate target and allowed alpha claims are recorded.
- [ ] WASB C3 status is recorded and reflected in alpha messaging.
- [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for alpha-plus.
- [ ] Owner records GO/NO-GO for the first alpha wave.

11. References

Internal docs:

- `docs/NEXT_STEPS.md`
- `docs/ROADMAP.md`
- `docs/data_pipeline/GOLDEN_VIDEOS.md`
- `docs/R2_EVAL_METRIC_DESIGN.md`
- `docs/RALLY_DETECTION_FIXES_PLAN.md`
- `docs/COMPUTE_DECOUPLED_SERVING/README.md`
- sibling `khelsutra/NEXT_STEPS.md`
- sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`
- sibling `khelsutra/docs/PER_RALLY_SELECTION.md`
- sibling `sports-data-collector/data/golden_source_videos.json`

Local verification commands used for the 2026-07-03 snapshot:

```powershell
python -m backend.eval.fusion_compare --features-dir output
python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>
python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor eval_baselines\heuristic_lovo_n6.json --version analysis-n15 --out <temp-out>
python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir output --no-report
python -m backend.eval.ablation --features-dir output --granularity group
```

Changelog

- 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`, golden-corpus docs, local eval artifacts, and the current 15-video corpus state.

10. assistant (2026-07-03T16:40:37.940Z)

Now let me check the archives/decisions directory to understand the ADR convention:

11. user (2026-07-03T16:40:38.626Z)

Architecture Decision Records

ADR-001: Precise shuttle tracking via self-hosted TrackNetV4/WASB on WSL2

Date: 2026-05-25
Status: Accepted

Context

The pipeline uses an AI multimodal provider (Gemini) for *semantic* rally segmentation. We need **precise frame-level detection of the moving shuttlecock** (and, generically, balls for other sports) to produce higher-quality candidate rallies and a fine-tuning signal.

We evaluated whether a commercially available model/API could do this. Findings:

- **Multimodal LLM APIs (Gemini, Claude, GPT-class) are the wrong tool for precise localization.** They reason well about *what* is happening (rally vs. dead time) but cannot reliably track a tiny, fast (~up to 400 km/h), motion-blurred shuttle frame by frame.
- The precision leaders are **purpose-built trajectory models**, which are open source and self-hosted, not big-vendor APIs:
  - **TrackNetV4** (TensorFlow) ? badminton-specific, heatmap + motion-attention, highest reported shuttle accuracy; ships `src/predict.py` producing per-frame (X, Y) coordinate CSV. https://github.com/AR4152/TrackNetV4
  - **WASB** (PyTorch) ? multi-sport strong baseline (badminton/tennis/volleyball/?), fits our `SportProfile` abstraction; needs a custom-video inference wrapper. https://github.com/nttcom/WASB-SBDT
- True turnkey commercial APIs (Roboflow hosted models) exist but cap below a tuned TrackNet; enterprise systems (Hawk-Eye, SkillCorner, TRACAB) are not API-accessible.

Decision

1. Adopt **self-hosted TrackNetV4** as the primary precise shuttle detector, with **WASB** as the multi-sport fallback.
2. Run these models under **WSL2 (Ubuntu)**, **not** native Ubuntu. The entire indexer already runs on Windows; WSL2 keeps a single dev environment and quarantines the TensorFlow/PyTorch dependencies. Native Ubuntu would require dual-boot and migrating the whole project for marginal gain.
3. Integrate via a **subprocess adapter**: a detector module that shells out to the WSL `predict.py`, reads the coordinate CSV, and converts trajectory ? candidate rallies feeding the existing stitching flow. This keeps the TF dependency fully isolated from the main Python environment.

Consequences

- **WSL I/O caveat:** `/mnt/c` access is slow. Stage match videos inside the WSL filesystem (e.g. `~/clips/`) before inference rather than reading large mp4s across the Windows?WSL boundary.
- TrackNetV4 (TensorFlow) and WASB (PyTorch) live in **separate conda envs**.
- Next step after integration: manual video annotation ? fine-tune the detector.

ADR-002: Use the sports-data-collector's rally annotations as a golden eval + temporal-segmenter training set

Date: 2026-05-26
Status: Accepted

Context

A sibling project, **sports-data-collector**, curates sports videos and stores per-rally `[start, end)` timestamps (plus stroke counts, scores) in an SQLite DB, with a commercial-license flag per source. We asked whether this data could improve the indexer's rally detection, and for which task.

Key findings:

- The annotations are **temporal-only** (when rallies start/end). They do **not**

contain
per-frame shuttle `(x, y)` coordinates, so they **cannot train the shuttle detector**
(TrackNetV4/WASB ? see ADR-001), which needs coordinate labels.
- They are an excellent **golden evaluation set** and a **training signal for the
temporal rally segmenter** (active vs. dead play over time) ? the stage that is today
a hand-tuned heuristic, not a learned model.
- The collector's five sports tables (`badminton_rallies`, `tennis_points`,
`cricket_deliveries`, `pickleball_rallies`, `tabletennis_rallies`) all share a
`(video_id, ordinal, start_time, end_time)` shape, and the indexer already has
`sport_registry` / `segmenter_registry`. So a *generic* `(start, end)` bridge serves
every sport.
- The two repos key videos differently: the collector by a human `video_id`, the indexer
by **file MD5**. A re-encode/re-download changes the bytes, hence the MD5.

Decision

1. Treat the collector's rally timestamps as (a) the **golden eval set** for rally
segmentation and (b) the **labels** for training a temporal segmenter ? explicitly
not as shuttle-detector training data.
2. Bridge the repos with a sport-agnostic **export** (`curate export` in the collector):
per-video `start,end` CSVs (the indexer's eval format) + a `manifest.json` that pairs
each CSV with its video file. Gate on `is_commercial` so only commercially usable data
flows through.
3. Honour the **MD5 keying** contract: evaluation/processing must run on the exact video
file recorded in the manifest; acquisition (ADR-003) therefore precedes processing.

Consequences

- The collector becomes the *producer* of labels; the indexer the *consumer* (processing +
scoring + training). Neither takes a hard dependency on the other beyond the manifest.
- Scoring is free and deterministic (reads the DB; no API). See `docs/EVALUATION.md`.
- Only invest in frame-level coordinate labels (for the detector) **if** evaluation shows

12. assistant (2026-07-03T16:40:41.120Z)

Let me check for any existing conventions about decision recording in DECISION_SNAPSHOT_REGRESSION.md or similar docs:

13. user (2026-07-03T16:40:41.689Z)

<!--
PROJECT DOC ? tracked design plan for a deterministic "quality-neutral proof".
Follows docs/PROJECT_DOC_TEMPLATE.md. §7 progress tracker is the source of truth.
-->

Decision-Snapshot Regression ? a deterministic quality-neutral proof

> **Status:** `DRAFT` ? **§2 decisions LOCKED (2026-06-25); ready to start §7 P1** · **Owner:** Avi · **Created:** 2026-06-25 · **Last updated:** 2026-06-25
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
> **Tracking anchors:** §7 progress tracker is the source of truth; indexed in `docs/README.md`; pointer in memory `current-state` once work starts.
> **Relation to existing docs:** **extends** `GOLDEN_REGRESSION_FIXTURES.md` (post-detector fixtures) and the existing characterization snapshot in `tests/test_golden_fixtures.py`; **peer of** the M4 quality-floor track and the GPU `scripts/nightly_reelcount.py` e2e nightly.
> **Honesty note:** claims tagged `[verified]` (checked vs code) / `[design]` (proposed).

---

0. TL;DR
A deterministic regression that proves a change is **quality-neutral** (behavior-preserving) by freezing and comparing the pipeline's **decision output** ? the rally segment list **+ the stitch/concat plan** ? run off a **cached trajectory** (GPU-free). It **extends** the existing characterization snapshot (`test_replay_matches_committed_expected`, today segmentation-only) down to the stitch plan, runs offline in CI, and gives the **behavioral proof** that lets a *genuine* refactor (not just `ruff format`) be confidently called quality-neutral. **Most important caveat: never hash the encoded reel `.mp4`** ? ffmpeg/x264/NVENC/mux-timestamps make those bytes non-deterministic, so a reel-MD5 would false-alarm on every ffmpeg bump.

1. Problem & goal
"Presumably quality-neutral" changes (refactors) need a cheap, deterministic proof that they actually preserve behavior ? so they can be merged (and auto-signed-off) with confidence. Today we have two weaker tools: **AST-identity** [verified] (used to bless `ruff format` in #379) only proves *formatting* is unchanged ? it cannot bless a real refactor (function extraction, rename, loop reflow) that changes the AST but should preserve behavior; and the full **real-video F1 / reel-count e2e** needs GPU + the gitignored corpus (`scripts/nightly_reelcount.py`) [verified] so it can't gate a PR. The decision-snapshot fills the gap: a deterministic, offline, behavior-level proof.

What "good" looks like: a refactor PR runs the snapshot in CI; byte-identical decision output ? provably quality-neutral ? eligible for auto-sign-off; a behavior-changing edit fails with a **readable diff**.

Read first: `tests/test_golden_fixtures.py::test_replay_matches_committed_expected` [verified] (the existing snapshot ? its own message: *"behaviour changed vs frozen snapshot (characterization, not correctness)"*), `backend/eval/golden_fixtures.py`, `backend/pipeline/stitcher/compiler.py` (the concat plan), audit doc **B1** (`motion`'s random fields), `scripts/nightly_reelcount.py` (the distinct GPU e2e).

2. Decisions *(D1?D11 **LOCKED** 2026-06-25)*

#	Decision (recommendation)	Rationale	Status
D1	**Snapshot the DECISION output, never the encoded reel.** Freeze the rally segment list + the stitch/concat plan; never MD5 the reel `.mp4`.	Encoded mp4 bytes are non-deterministic (ffmpeg version string, x264 threading, NVENC, mux timestamps) ? a reel-MD5 false-alarms on every ffmpeg bump/machine. The repo's own `nightly_reelcount.py` already asserts count+metrics, **not** bytes [verified].	**LOCKED** (2026-06-25)
D2	**Input = a cached trajectory (detector output), not raw video.** Run from a pinned trajectory CSV (the `fx_r0X` fixtures).	The WASB detector (video?trajectory) is the GPU/WSL + CV-float-non-deterministic upstream; caching its output makes the proxy deterministic + CI-runnable [verified ? the fixtures are post-detector].	**LOCKED** (2026-06-25)
D3	**Coverage = segmentation decision + stitch/concat plan; STOP before the encode.** Extend the existing segmentation snapshot down to the concat manifest (clips, cut points, order, padding).	Segmentation is already snapshotted [verified]; the stitch plan is the deterministic missing layer. The encode is non-deterministic (D1).	**LOCKED** (2026-06-25)
D4	**Compare = canonical-JSON snapshot with a readable diff** (an MD5 of the canonical JSON may be a fast-path summary, but the field-level diff is the artifact).	A refactor that trips it needs a debuggable diff, not "hash changed" ? mirrors `test_replay_matches_committed_expected`'s mismatch report [verified].	**LOCKED** (2026-06-25)
D5	**Snapshot only deterministic segmenters/fields; exclude the random fabricated fields.** Use a deterministic segmenter (CONTROL/fusion on trajectory) and a `{start,end,shuttle}`-style schema that excludes `motion`'s random `server/receiver/shot_count/avg_speed`.	`motion` fabricates random fields (audit **B1**) [verified] ? non-snapshot-able as-is. **B1's fix (drop the fabrication) is the enabler that would make `motion` snapshot-able.**	**LOCKED** (2026-06-25)
D6	**The reel mp4 gets a FUNCTIONAL smoke, not a byte hash** (exists, ~duration within tol, N segments, sane streams).	Corollary of D1 ? verify the encode is *sane*, not *identical*.	**LOCKED** (2026-06-25)
D7	**Runs in per-PR CI, offline, CONTROL-only ? a refactor gate, not a quality

floor.** | The point is catching a behavior change in a refactor *before merge*; it must be fast, GPU-free, on the pinned CONTROL config. | **LOCKED** (2026-06-25) |
D8	**Intentional changes re-freeze WITH a documented rationale (Launch Report).** A `--update-snapshot` flag exists, but a mismatch on an *intended* change must be re-frozen with the Launch-Report rationale recorded.	Without the discipline a snapshot becomes a rubber stamp; ties to the launch-report convention.	**LOCKED** (2026-06-25)
D9	**Scope boundary: this is the quality-NEUTRAL proxy ? distinct from the M4 quality-FLOOR (per-slug F1), Tier-2 real-video e2e (GPU), and `nightly_reelcount.py`.** Keep them separate + complementary.	Conflating "behavior unchanged" with "quality is good enough" muddies both.	**LOCKED** (2026-06-25)
D10	**Deterministic-segmenters-only, ENFORCED by a determinism guard.** Cover only deterministic segmenters; to *ensure* no non-deterministic one slips in (? a flaky snapshot), the harness runs each fixture **twice and asserts byte-identical decision output** before freezing ? any segmenter whose two runs differ (hidden `random`, set-ordering, float/time non-determinism) is **rejected from snapshot scope** and flagged. Mechanical, not a hand-maintained allowlist.	Owner, 2026-06-25. `motion` is non-deterministic until **B1 / #385** lands; then it auto-becomes eligible (no allowlist edit).	**LOCKED** (2026-06-25)
D11	**Auto-sign-off integration is STAGED: ADVISORY first ? GATING later.** Ship the snapshot as an advisory/informational check; graduate it to a **gating** check (gates a non-AST-identical refactor's auto-LGTM) only after it's proven stable on real refactors.	Owner, 2026-06-25. Resolves §8 Q1 ? the strategic payoff, de-risked by starting advisory.	**LOCKED** (2026-06-25)

4. Design / architecture
Flow (all offline, deterministic):
`pinned trajectory CSV ? [segmentation: CONTROL segmenter] ? segment list ? [deterministic stitch-PLAN builder: clips + cut points + padding + order, NO ffmpeg] ? canonical JSON {schema_version, segments[], stitch_plan{}} ? MD5 / field-diff vs frozen golden`.
Mismatch ? behavior changed (decide if intended; re-freeze via D8).

Reuse map [verified unless noted]:
- `tests/test_golden_fixtures.py::test_replay_matches_committed_expected` + `backend/eval/golden_fixtures.py` ? the existing characterization snapshot at the **segmentation** layer; extend its `expected` schema + replay to also carry the stitch plan. **(reuse + extend)**
- `eval_baselines/fixtures/` (`fx_r0X` + `manifest.json`) ? the pinned post-detector trajectories. **(reuse)**
- `backend/pipeline/stitcher/compiler.py` ? builds the ffmpeg concat today; **[design]** factor out a pure `build_stitch_plan(segments, cfg) -> dict` (the *decision*) from the *encode*, so the plan is hashable without running ffmpeg. **(new seam)**
- `eval_baselines/` snapshot store + the cross-OS enforcement already used by the golden suite. **(reuse)**

6. Honest limits ? what this does NOT do
- Does **not** prove *quality is good* ? only that *behavior is unchanged* vs the frozen golden (that's M4 / F1-floor's job).
- Does **not** cover the **encode** (pixels/bytes) ? only the decision + stitch plan; the reel gets a functional smoke only (D6).
- Does **not** run the **real detector** (GPU/WSL) or real videos ? it runs from cached trajectories (Tier-2 / `nightly_reelcount.py` cover that, GPU-gated).
- A **green snapshot does not bless a behavior change** ? it only certifies *no* change; an intended change must be re-frozen with a rationale (D8).
- Only as good as the **fixtures' coverage** ? pipeline paths not exercised by `fx_r0X` aren't protected.

7. Deliverables & progress tracker ? **source of truth**
Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One small PR per row, branched from `origin/master`.

ID	Deliverable	Depends on	Gated?	Status	PR
P0	This doc ? decisions in §2 locked	?	Owner	?	(this PR)
P1	Canonical decision-snapshot schema + deterministic serializer (segments + stitch plan; excludes non-det fields)	P0	No	?	?
P2	Factor `build_stitch_plan()` out of `compiler.py` (pure, ffmpeg-free) + unit test

```
| P0 | No | ? | ? |
68      | P3 | Extend `golden_fixtures` / `test_golden_fixtures` to capture + compare the stitch
plan **+ the D10 determinism guard (run each fixture twice ? assert byte-identical; reject non-
deterministic segmenters)** | P1, P2 | No | ? | ? |
69      | P4 | Freeze CONTROL snapshots for the existing `fx_r0X` fixtures | P3 | No | ? | ? |
70      | P5 | `--update-snapshot` flow + Launch-Report hook (D8) | P3 | No | ? | ? |
71      | P6 | Wire the snapshot as a per-PR CI characterization gate (CONTROL-only) | P4 | No |
? | ? |
72      | P7 | **Surface the snapshot on refactor PRs ? ADVISORY** (informational; does not
gate) ? per D11 | P6 | No | ? | ? |
73      | P8 | **(gated)** Graduate to GATING auto-sign-off: a snapshot-match marks a non-AST-
identical refactor quality-neutral ? LGTM-eligible (only after P7 proven stable) ? per D11 | P7
| **Owner** | ? | ? |
74
75      ## 8. Open questions ? owner
76      1. ? **RESOLVED (D11):** advisory first ? graduate to gating. *(The strategic payoff,
de-risked by staging.)*
77      2. ? **RESOLVED (D10):** deterministic-segmenters-only, **enforced by the double-run
determinism guard**; **B1 is an enabler, not a blocker** ? and it's being fixed in **#385**,
which makes `motion` auto-eligible.
78      3. **(still open) Fixture scope:** reuse the existing `fx_r0X` (which stop at
segmentation) by adding a stitch-plan `expected`, or add a dedicated fixture set?
79      4. ? **DONE: D1?D11 LOCKED 2026-06-25.**
80
```

14. assistant (2026-07-03T16:40:43.785Z)

```
Now let me check what changes were made to DOC_STATUS.md and NEXT_STEPS.md in this PR:
```

15. user (2026-07-03T16:40:45.201Z)

```
diff --git a/docs/DOC_STATUS.md b/docs/DOC_STATUS.md
index 34411fd..2541672 100644
--- a/docs/DOC_STATUS.md
+++ b/docs/DOC_STATUS.md
@@ -30,6 +30,7 @@ Before a foundational/tracked doc moves to `docs/archives/`:
 ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the completeness
source-of-truth)
 | Doc | Status | Archive-ready? | Next action |
 |---|---|---|---|
+| `ALPHA_LAUNCH_READINESS.md` | **DRAFT** ? cross-repo alpha readiness tracker created
2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-
model/commercial rally-detection gates | No | Execute §7 in order: corpus portability + n=15
baselines, hosted alpha serving, ablation/fusion reruns, then launch GO/NO-GO |
 | `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M9 LARGELY SHIPPED**
(#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0 weights
pushed (`weights/0.1.0-l4/`) | No | M11 flywheel + M12 gdrive-api (owner residual:
counsel/prices/OAuth/real-run within the D17 cap) |
 | `RALLY_QUALITY_RESEARCH.md` | Foundation, **tracked & ACTIVE** (§7) | No | Drives the
R-series; complete when §7.1 single quality number is attributed |
 | `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation
pending) | Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster |
diff --git a/docs/NEXT_STEPS.md b/docs/NEXT_STEPS.md
index f186ff9..a9f35a8 100644
--- a/docs/NEXT_STEPS.md
+++ b/docs/NEXT_STEPS.md
@@ -9,7 +9,7 @@

 ## ? Prioritized pending ? all tracks
 **P0 ? highest leverage / unblocks the most**
-0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings
lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only**
(`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/`
SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service**
(M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA
```

ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record: [[ui-driven-cloud-serving-gap]].
+0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only** (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service** (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record: [[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus review:** `docs/ALPHA_LAUNCH_READINESS.md` §7.
 0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-included app ? `docs/PACKAGING_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip install` ? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** **(1)** P3 **GPU validation + screencast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB 0-rally tuned-config blocker), **(2)** P2b-install (uv/script + truly-local default config), **(3)** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.
 1. ? **`COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged (default-OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, $0). All §8 gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a **$100/?10k cap** (D17).
 2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus training feed on it.

16. assistant (2026-07-03T16:40:48.611Z)

Now let me check if there are any existing cross-repo trackers or similar patterns:

17. user (2026-07-03T16:40:57.562Z)

./badminton-highlight-indexer/docs/ALPHA_LAUNCH_READINESS.md
./badminton-highlight-indexer/docs/archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
./badminton-highlight-indexer/docs/archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md
./badminton-highlight-indexer/docs/archives/past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md
./badminton-highlight-indexer/docs/BACKLOG.md
./badminton-highlight-indexer/docs/DOC_STATUS.md
./badminton-highlight-indexer/docs/NEXT_STEPS.md
./badminton-highlight-indexer/docs/PACKAGING_PLAN.md
./badminton-highlight-indexer/docs/POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
./badminton-highlight-indexer/docs/README.md

18. assistant (2026-07-03T16:40:59.628Z)

Let me check the sibling khelsutra repo docs structure:

19. user (2026-07-03T16:41:00.464Z)

```
total 192
drwxr-xr-x 1 avidu 197609     0 Jul  2 21:13 .
```

```
drwxr-xr-x 1 avidu 197609     0 Jul  2 21:13 ..
-rw-r--r-- 1 avidu 197609 17371 Jul  2 08:13 ANNOTATE_AND_TRAIN.md
drwxr-xr-x 1 avidu 197609     0 Jun 23 00:19 archives
-rw-r--r-- 1 avidu 197609 16474 Jul  2 21:13 DRY_RUN_CUJ.md
-rw-r--r-- 1 avidu 197609 37141 Jul  2 21:13 LOCAL_APPLIANCE_ARCHITECTURE.md
-rw-r--r-- 1 avidu 197609 17451 Jun 23 00:19 MULTILINGUAL_EXPANSION.md
-rw-r--r-- 1 avidu 197609 29326 Jun 22 15:27 PER_RALLY_SELECTION.md
-rw-r--r-- 1 avidu 197609  5000 Jul  2 21:13 PORTABILITY.md
-rw-r--r-- 1 avidu 197609  3790 Jul  2 08:13 README.md
-rw-r--r-- 1 avidu 197609  6997 Jun 22 15:27 REQUEST_ACCESS_FORM.md
-rw-r--r-- 1 avidu 197609 29193 Jul  2 08:13 STUDIO_MEDIA_LIFECYCLE.md
```

20. assistant (2026-07-03T16:41:05.378Z)

Let me check NEXT_STEPS.md to see if there's a convention about cross-repo coordination:

21. user (2026-07-03T16:41:06.154Z)

```
  font eyeball) CLEARED;** carried forward to `NEXT_STEPS.md`: a delta native-review of the
post-sign-off
  string additions, and **B7** (engine backend l10n, deferred).
- **`MULTILINGUAL_EXPANSION.md`** ? `IN PROGRESS`: **Phase A (es/da/id) shipped** ([#67]);
**Thai
  (B) / CJK (C) remain owner-gated**. Adds **7 locales** (th/id/da/es/zh-Hans/ja/ko) beyond the
  en/kn/hi foundation with en/kn/hi-level diligence; peer of the archived
  `FRIENDLY_AND_MULTILINGUAL.md`, reusing its foundation.
```

22. assistant (2026-07-03T16:41:08.292Z)

Let me check if there's a NEXT_STEPS.md in khelsutra:

23. user (2026-07-03T16:41:09.086Z)

```
-rw-r--r-- 1 avidu 197609 9773 Jul  2 21:13 NEXT_STEPS.md
```

24. user (2026-07-03T16:41:11.088Z)

```
1       # Next steps (live) ? KhelSutra brand / site
2
3       Prioritized "what to do next" for the **`khelsutra` product repo**. Refresh at each
checkpoint.
4
5       **Where we are (updated 2026-07-02):** `khelsutra.guru` is **LIVE and fully shipped** ?
landing page + `/capture`
6       checklist + OG image + a **segmented Request-Access form** (academy/individual + GPU +
storage + free
7       text). On **Cloudflare** (Worker `hidden-king-khelsutra` + D1 `khelsutra-access`): the
form persists to
8       D1 **and emails a summary to the owner on every registration** (Email Routing send
binding ? verified
9       live). **Cloudflare is the single canonical host** ? the DO droplet mirror
(`168.144.126.176`) was
10      **decommissioned + deleted 2026-06-25** (#117; data archived to
11      `gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/`). The
dependency-free
12      **Node + SQLite** host (`site/server.js`) remains VM-portable (live-proven on the
droplet before its
13      retirement) but nothing is deployed on it today; alpha serving is **paused pending the
Cloud Run move**.
14      **Git auto-deploy is connected** (Workers Builds, root `site/`): merge to `master` ?
deploys; **CI gates
15      every PR** ? three jobs: the full site `npm test` (Worker + VM host + assets + UX), the
`web/` SPA build
16      and coverage floor, and a real-browser layout check. Shipped across ~90 merged PRs (the
```

form, 6-language
localization, the per-rally MVP, VM portability). Docs: `docs/REQUEST_ACCESS_FORM.md`,
`docs/PORTABILITY.md`.

Next product threads (owner ideas, 2026-06-20)
- **? DONE ? Friendly & Multilingual (ARCHIVED 2026-06-22; human gates cleared):** the product is
  non-tech-friendly and **fully localized to all 6 languages** (en/kn/hi/es/da/id) across both surfaces
  (`web/` SPA + `site/` marketing) ? including the home page (hero, all sections, the `#join` access form,
  footer), the **Privacy Policy** page, **per-page `<title>`s**, and **es/da/id academy flyers** (with
  localized SPA screenshots). The responsive header is fixed and a **headless-Chromium CI suite** guards
  layout, cross-locale render (live DOM == catalog), and **accessibility (axe-core WCAG 2.1 AA)**. Full
  record: **`docs/archives/FRIENDLY_AND_MULTILINGUAL.md`**. Post-archive PRs: #86 (own-model) · #87 (es/da/id
  site) · #105 (form+footer + render guard) · #106 (Privacy page) · #107 (a11y fixes + axe guard + page
  titles) · #109 (flyers). **Owner gates CLEARED:** native-speaker review of the AI-drafted catalogs ? +
  real-device font-glyph eyeball ?. **Remaining (small/owner):**
  - **Delta native-review** of the strings ADDED after the owner's sign-off (#105/#106/#107/#109 ? the
    form/footer/privacy/title/flyer copy is AI-drafted): a small top-up review before a wide public launch.
  - **B7 (deferred, not a v1 blocker):** engine backend customer-facing l10n (report `customer_*` +
    rejection reasons) ? cross-repo Python (`badminton-highlight-indexer`), owner-gated **D-BACKEND-DEPTH**.
- **Marketing video on the site** ? embed a short walkthrough of the record ? index ? reel process,
  themed for **serious enthusiasts**, on the landing page. ? **A real-UI CUJ screencast now exists**
  (Playwright recording of the per-rally SPA driven against real GPU-run data ? `promo_cuj.mp4`/`.gif`
  in `gs://khelsutra-rally-corpus/promo/`). Decide: use it as the marketing clip (? shows real court
  footage + a dev "P5" badge ? clean up before public embed), and/or produce a polished Grok/Gemini cut.
- **Interactive per-rally selection** (big product feature) ? ? **MVP / M0 COMPLETE** (the full
  pick ? query ? select ? compile ? download loop on existing engine endpoints). Lives in the alpha SPA
  (`web/`). **? Tracked design + P-tracker (source of truth): `docs/PER_RALLY_SELECTION.md` (IN PROGRESS;
  owner-approved, D8?D11 locked).** **Done: P0 design (#11), P1 scaffold (#14), P2 grid+hook (#21), P3
  `parseQuery()` query bar (#28), P4 SelectionFooter + `min_shot_count` warning (#30), P5 real load +
  Build?compile?preview?download (#31).** Verified e2e vs a stub engine + a CUJ replay against the real
  L4-run rallies + reel. **NEXT (owner-gated, engine repo): M1?M5** ? `GET /api/health` + `download_url`
  (P6), RESTful `?/rallies` alias (P7), multi-video home + auth (P8), per-rally preview/timeline (P9),
  corpus correction loop (P10), localized NL (M5). **Optional:** a fresh live `/api/process` GPU
  inference once the engine session's M2 training frees the single GPU.

Queued (site polish)
- **Always-Use-HTTPS + HSTS** (Cloudflare SSL/TLS) ? closes the plain-HTTP flag.
- **Cloudflare Web Analytics (owner-gated; FREE)** ? code side DONE (the dead

placeholder beacon was
  removed from `site/public/index.html`; analytics is now via **Automatic Setup**, no in-source token).
  Owner step (needs CF login): the site is a Cloudflare **Worker** (not Pages) on the proxied
  `khelsutra.guru` zone ? enable via **Analytics & Logs ? Web Analytics ? Add a site ? `khelsutra.guru`**
  (Automatic Setup is default-on for proxied hostnames; edge-injects the beacon on every page). Then
  filter the view to `hostname = khelsutra.guru` to exclude `*.pages.dev`/preview noise, and decide
  separately whether `app.khelsutra.guru` (the SPA) needs its own enablement (history-router only).
  Verify (two gotchas): the Worker's asset responses must NOT send `Cache-Control: ?, no-transform`
  (Cloudflare can't rewrite that ? no injection), and after a deploy the rendered HTML should contain the
  edge-injected beacon. **DONE WHEN:** the Web Analytics dashboard shows incrementing page views for
  `khelsutra.guru`.
- **Any future VM ? production** *(re-scoped 2026-07-02 ? was "Droplet ? production"; the droplet was
  deleted 2026-06-25, #117)* ? only if a VM standby is ever wanted again: `site/scripts/provision-vm.sh`
  is provider-agnostic; add TLS + a subdomain (`vm.khelsutra.guru` via Caddy) and **VM-path SMTP** so the
  Node host also emails (currently `noopNotify`). Not urgent ? Cloudflare is the single canonical host.
- **Revoke the droplet SSH key (`~/.ssh/khelsutra_droplet`) ? still actionable, do it now (owner):**
  the droplet is gone (deleted 2026-06-25) but the local keypair lingers; revoke/delete it rather than
  waiting on any VM work.
- **DMARC** TXT for deliverability; delete any orphan `khelsutra-site` Worker; clean CF D1 test rows
  (`deploy-test@`, `email-test@`).

Bigger arcs (design-first)
- **? Studio media lifecycle (ingest ? store ? annotate ? train)** ? **Tracked project: `docs/STUDIO_MEDIA_LIFECYCLE.md`
  (IN PROGRESS ? storage half COMPLETE 2026-07-01).** The seamless flow: add a video ? store it in a user-chosen medium
  (S3/GCS/Drive/local) ? free-space checked ? annotate in-UI ? contribute to the segmenter. **SHIPPED 2026-07-01:**
  free-space guards (vault + engine, P1/P2), the `/api/capabilities` storage block (P3), `POST /api/assets`
  store-into-medium (P4), the SPA medium picker (P5), the "my videos" library (P6), and in-UI annotation (P7?P9).
  Remaining = owner-gated only: P10 (corpus promotion) + P11 (train/re-eval ?F1) + the `L` beta follow-ups.
  §7 tracker is the source of truth.
- **Single-box appliance (KhelSutra Studio)** ? **? Tracked design: `docs/LOCAL_APPLIANCE_ARCHITECTURE.md` (DRAFT, owner-gated).**
  The `web/` SPA drives the full **ingest (Drive / bucket URI / local upload) ? index ? pick rallies ? reel ? download**
  loop on ONE box, behind an **edge auth gate** (engine stays `127.0.0.1`, CORS `*` is moot). It is the single-box
  profile of the engine's decoupled `StorageBackend` + `backend.worker` architecture. MVP ran on the CPU droplet
  (retired 2026-06-25) via **Gemini/stub**; **real local-GPU detection needs a GPU host** (any VM provider / Cloud
  Run+GPU ? owner priority ASAP, runs parallel to the UI MVP). Per-rally selection = one screen. Owner decisions pending in §10 (edge auth,
  retention, cost ceiling, GPU host, P5 bridge).
- **Alpha SPA** in `web/` (Vite + React + TS + Tailwind) ? hosts the operator console +

the per-rally selection UI above.
90   - **BYO-storage + autosync** ? connect Drive/OneDrive/GCS/S3, watch the folder, auto-prepare reels
91   (engine `PLATFORM_ARCHITECTURE.md §3`). The form's GPU/storage answers are the demand signal.
92   - **Control plane** (`control-plane/`) at beta.
93
94   ## Org & repos
95   - ? **DONE (2026-06-20): proprietary repos consolidated into the `Khelsutra` org (private).** Transferred
96   `badminton-highlight-indexer` (engine), `sports-data-collector`, `sports-obsreport` ? `Khelsutra/*` (stay private; no
97   open PRs/forks; GitHub redirects git ops so active sessions kept working; local remotes repointed + verified).
98   `rally-annotator` stays in the org **PUBLIC** (OSS MIT plugin the site links to ? do not make private). Owner is org
99   admin (full access) and can **invite a contributor** (org member / per-repo outside collaborator) anytime.
100   - ? **`khelsutra` (this site repo) ? owner step pending.** A transfer breaks the **Cloudflare Workers-Builds** git
101   connection (auto-deploy) until the Cloudflare GitHub App is installed on the `Khelsutra` org and the Worker
102   reconnected (dashboard). When ready: install the CF app on the org ? `gh api repos/avidullu/khelsutra/transfer -f
103   new_owner=Khelsutra` ? reconnect the Worker to `Khelsutra/khelsutra` ? verify a deploy.
104   - ? **Engine CI under the org** ? verify the engine's GitHub Actions run on the next PR (free orgs default Actions-on;
105   enable in org settings if a run doesn't start).
106
107   ## Convention
108   Substantial work gets a tracked doc in `docs/`; one PR per item off `origin/master`; merge to `master`
109   auto-deploys (CI must be green).
110

25. assistant (2026-07-03T16:41:13.446Z)

Now let me check the actual README.md to see if there's guidance on where trackers should live:

26. user (2026-07-03T16:41:13.970Z)

1   - RALLY_QUALITY_RESEARCH.md ? **? Foundation doc** for closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps, the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an **adversarially-verified, cited external-research synthesis** (TAL/TAS boundary heads, racket-sports rally-state cues, small-data, eval rigor), a **prioritized cheap?durable roadmap with falsifiable success criteria**, and (§7) a **tracked project plan** ? work-item checklist, sequencing, the single quality **number** attributed to the effort, and a **definition of done** (project completes when all tiers land + the number is recorded). The plan that the QUALITY_ITERATIONS.md log executes against; ties together EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS / OWNED_MODEL_IMPLEMENTATION_PLAN.
2   - RALLY_DETECTION_GAP_CLOSURE.md ? **Tracked execution doc (2026-06-18)** for closing the remaining rally-detection accuracy gap **per video, before launch**, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it: boundaries are tight (R2/R3/R4 holds) ? the gap is **detection**: (G1) recall (local & Gemini both miss ~half on hard clips, and *different* rallies ? complementary), (G2) precision (local over-fires), (G3) **rally-END over-extension on the quick-return-after-point** (shuttle still moving ? needs a *player-state* signal). Levers (owner-gated): player-kinematics + shuttle-in-hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally_gate` Gate A wiring, complementary fusion. **Peer** of `RALLY_QUALITY_RESEARCH.md` (execution tracker, not research); §7 is the source of truth.

- PROMINENT_COURT_DETECTION.md ? **Design + experiment (2026-06-19, default-OFF, owner-gated flip)** for the **multicourt G2 over-fire**: identify the **prominent (foreground) court** (the one covering the majority of the frame) and keep only rallies whose shuttle lives inside it ? re-defining the shuttle signal `(x,y,visible)` ? `(x,y,visible-AND-inside-prominent-court)` *before* windowing, so a background/adjacent-court shuttle never forms a rally. 2D floor polygon (MVP) ? **pseudo-3D play-volume** (refinement, so high clears aren't clipped). Born from the GX010142 rally-#1 background-court FP; reuses `fusion_features.point_in_polygon` + `FusionConfig.court_polygon`. §8 progress tracker is the source of truth.
- RALLY_DETECTION_QUALITY_REPORT.md ? **Technical-paper checkpoint (2026-06-17)** of rally-detection quality: the honest measured state (over-segmentation milestone tolerance-F1 0.091?0.323 at n=11, p=0.002; merge-rate 0.694?0.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY_QUALITY_RESEARCH.md` + `QUALITY_ITERATIONS.md`.
- EVALUATION.md ? How we measure and improve quality.
- EXPERIMENT_HARNESS.md ? Design (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with **zero production regression** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.
- QUALITY_ITERATIONS.md ? **Living log** of rally-detection quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-chronological record; terminal-state snapshots get archived under `archives/quality-iterations/`.
- ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a **signal-fusion framework** for rally detection ? producers emit confidence-scored **signals** at three temporal **scopes** (frame detectors / window analyzers / session analyzers), a learned **scorer** weights them, and an auditable **report-first reconciliation** pass lints the decision against the evidence. The spine: **no special-casing in the decision path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT_HARNESS.md` and `MULTI_SIGNAL_FUSION_PLAN.md` (ADR-007 modular rally layer).
- PERSONALIZATION_PLAN.md ? Design (owner-adopted **hybrid**; Phase-0 landing) for the **per-user personalization feature on a generalization-first baseline**: a thin per-user *tuning* layer gated by the honest small-N statistics (a **min-N promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; *"a tool that gets better the more you help it"*). Records the **locked owner decisions** (identity, storage, `min_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).

- GOLDEN_REGRESSION_FIXTURES.md ? **Design (owner-gated, NOT started):** video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **§7 progress tracker + definition-of-done**. Extends [GOLDEN_DATA_SHARING.md](data_pipeline/GOLDEN_DATA_SHARING.md).

? Documentation status & governance
- [AUDIT_REPORT_khelsutra-guru_2026-07-02 (? COMPLETE / ARCHIVED)](archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md) ? the five-repo `khelsutra-guru` workspace code audit + its folded-in **Remediation tracker** (P0?P28 all closed). **Archived 2026-07-02** once every remediation wave landed and the last residuals were closed or filed as issues (#443 CORS, #444 corpus Phase-2, #445 GCS eval; + #301/#327/#332/#384). Absorbed the prior `CODE_CLEANUP_AUDIT_2026-06-24` + June-2026 hardening sweep (both archived).
- DOC_STATUS.md ? **? living Documentation Status & Archival Register**: every foundational/tracked doc's lifecycle + archive-readiness, the **archival due-diligence checklist** (*honestly update ? then archive*), and the open doc-hygiene findings. Review here; update rows as docs complete.

Product / platform plans

- ALPHA_LAUNCH_READINESS.md ? **? Active tracked project (`DRAFT`, created 2026-07-03):** cross-repo alpha readiness tracker after the golden corpus reached 15 videos. Separates the hosted product-alpha lane (shareable upload?process?pick?reel) from the owned-model/commercial rally-detection gate (ship target + WASB-C3 still unresolved). §7 is the source of truth for the concrete next PR sequence: corpus portability, n=15 baseline refresh, ablation/fusion reruns, hosted serving, alpha copy/launch review.
- [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? **? Active tracked project (2026-06-21, `DRAFT` owner-gated):** the productionization successor to DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend` (what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU, in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI). Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY proof for $0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a [runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). §7 M0?M10 tracker is the source of truth.
- [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ? **DELIVERED + ARCHIVED (2026-06-21).** Lifted the rally pipeline onto a **rented cloud-native GPU** doing bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB runner**; **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ? DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving" subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under `docs/archives/past_projects/DECOUPLED_COMPUTE/` (incl. `DEPLOY_REPORT_GCP_2026-06-20.md`).
- PACKAGING_PLAN.md ? **? Active tracked project (`IN PROGRESS`, 11 PRs merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ? `pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra SPA** single-origin, **torch-free** (LIGHT tier), cross-platform, no native component (decision D10/O1). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP_APP_PLAN.md` narrative.** §7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). **? Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.
- DESKTOP_APP_PLAN.md ? **? Superseded narrative (by `PACKAGING_PLAN.md`; D10 = self-hosted webserver, NOT a desktop app).** Original scoping doc (2026-06-18, `[design]`): package the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU* enthusiast laptop (WASB is ~3.4× real-time on a laptop 2070S)? §7 tracker = **P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **`LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

Practical / ongoing

- CREATING_REELS.md ? **? End-user guide:** install the LIGHT-tier app ? `indexer --app` ? process a video ? pick rallies ? download a watermarked reel. Task-first usage walkthrough (web UI + the API chain), with a §6 pointer to the cloud compute option. LIGHT tier only (offline `motion` / optional Gemini; no on-device GPU or one-click cloud). Install mechanics ? root [`README.md`](../README.md); packaging roadmap ? [`PACKAGING_PLAN.md`](PACKAGING_PLAN.md).
- CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md ? **? Operator runbook:** `gs://` bucket videos ? highlight reels on the approved **L4 Cloud Run** (scale-to-zero, $0 idle). Copy-pasteable: build image ? create the GPU Job ? cloud-serving control plane ? process(cloud)?query?compile?download ? prove the path (`verify_compute_path.py`) ? $0 teardown. **Verified e2e 2026-06-23.** Raw deploy knobs ? [`../deploy/cloudrun/README.md`](../deploy/cloudrun/README.md).
- SETUP_NEW_MACHINE.md ? **? The canonical "spin up a box ?

ready for training & inference" runbook.** Copy-pasteable, human-followable, **verified end-to-end on DigitalOcean (2026-06-20)** ? but **compute is now GCP-only (ADR-013)**, see [`deploy/gcp/README.md`](../deploy/gcp/README.md): credentials ? provision ? bootstrap ? suite (922 passed) ? secret-free CPU inference (no GPU/keys) ? real Gemini inference ? Gen-0 training, with pointers to the GPU/native-detector + bucket paths. The `deploy/digitalocean/*` scripts are provider-agnostic and reused on GCP.
- DATA_IN_GCS.md ? **? The corpus source-of-truth map: where every piece of data lives in `gs://khelsutra-rally-corpus`** (canonical layout · current contents + drift · how `backend.worker` consumes it · the still-local-only **gap** · migration plan). Read this so a session pulls data from the bucket, not from someone's `C:`/`F:`/Drive ? the goal is to remove the local box as a blocker. Pairs with the two runbooks below.
- [`deploy/gcp/nightly_regression/README.md`](../deploy/gcp/nightly_regression/README.md) ? **? Nightly engine-regression OPERATIONS MANUAL (one-stop shop):** enable · run on demand · change any setting (email recipients/frequency/clips/config) · observe · pause/disable/remove. The full WASB-pipeline reel-count+cost+email nightly on an ephemeral GCP GPU VM (design/status: [`NIGHTLY_ENGINE_REGRESSION.md`](NIGHTLY_ENGINE_REGRESSION.md)). The lighter CPU cached-trajectory tripwire is `scripts/nightly_regression.py` (local Windows Scheduled Task, PR #202).
- TRACKNET_WSL_SETUP.md
- Evaluation harness and calibration drivers (in `backend/eval/`)

Data layer (collection · annotation · golden sets ? **no ML**)
- [`data_pipeline/`](data_pipeline/README.md) ? **the isolated data pillar**: how footage is recorded (`VIDEO_RECORDING_GUIDELINES`), golden sets are created + which videos (`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS`), and data is shared/ingested (`GOLDEN_DATA_SHARING`). Deliberately **no segmentation/stitching/model** detail ? that's the ML pillar (top level + `COMPUTE_DECOUPLED_SERVING/`). The **annotation-vendor (Roora)** engagement is archived under `archives/roora-vendor/`.

Conventions & templates
- PROJECT_DOC_TEMPLATE.md ? Reusable skeleton for a **tracked project doc**: status header · decisions-locked · progress tracker (?/?/?) · definition-of-done · archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7 and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.

Historical documents
Completed plans, old research, ADRs, and checkpoints live in [`archives/`](archives/).

- **June-2026 test-hardening sweep** ? **COMPLETED** (2026-06-14; T1?T5 + full loop + verification per #162) and **archived**; the former top-level pointer-stubs were removed 2026-07-02 (absorbed into the umbrella's *Prior audits*). Full records: [`archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md`](archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md), [`archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md`](archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md), [`archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md`](archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md).
- **CODE_CLEANUP_AUDIT (2026-06-24)** ? the prior deep engine audit (35 findings; Phase 1 complete, Phase 2 largely shipped, remaining = the `main.py` god-module split). **Absorbed into the umbrella** (see its *Prior audits (absorbed)*) and archived ? [`archives/past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md`](archives/past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md).
- **KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN** ? the five-repo audit execution board, **folded into the umbrella** as its *Remediation tracker (live)* and archived ? [`archives/past_projects/KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md`](archives/past_projects/KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md).
- DECISION_SNAPSHOT_REGRESSION.md ? **? Tracked project (decisions D1?D11 LOCKED 2026-06-25, #383; implementation P1?P7 pending):** a deterministic **quality-neutral proof** ? freeze + compare the pipeline's **decision output** (rally segments + stitch/concat plan) off a cached trajectory, GPU-free, in CI. The behavioral proof that lets a *genuine* (non-AST-identical) refactor be confidently called quality-neutral. **Never hashes the encoded reel** (non-deterministic). Extends the existing characterization snapshot (`test_golden_fixtures.py`) from segmentation to the stitch plan; peer of the M4 quality-floor + GPU `nightly_reelcount.py`. §7 tracker is the source of truth.

47   - **Field-ops / customer-feedback** (recruiting flyer · field-visit playbook · PMF) ? [`archives/field-pmf/`](archives/field-pmf/) (archived 2026-06-21; PMF validation still an active *business* thread ? see `NEXT_STEPS.md`).
48   - **Roora annotation-vendor** engagement (labeling brief · vendor bake-off) ? [`archives/roora-vendor/`](archives/roora-vendor/) (archived 2026-06-21).
49   - Superseded: per-video output layout ? [`archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md`](archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md) (? `PER_VIDEO_WORKSPACE.md`).
50
51   See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md` for classification of small tech debt / starter projects.
52

27. assistant (2026-07-03T16:41:16.211Z)

Now let me analyze the key convention points. Let me check if DOC_STATUS.md references a convention about where cross-repo work is tracked:

28. user (2026-07-03T16:41:17.559Z)

21:3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`, `docs/NEXT_STEPS.md`, and any cross-doc links ? the new path + status.
33:| `ALPHA_LAUNCH_READINESS.md` | **DRAFT** ? cross-repo alpha readiness tracker created 2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-model/commercial rally-detection gates | No | Execute §7 in order: corpus portability + n=15 baselines, hosted alpha serving, ablation/fusion reruns, then launch GO/NO-GO |
78:`PROJECT_DOC_TEMPLATE.md` · `NEXT_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` · `EVALUATION.md` · `ABLATION_LAYER.md` · `SETUP_NEW_MACHINE.md` · **`CREATING_REELS.md`** (? end-user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) · **`CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md`** (? operator runbook ? `gs://` videos ? reels on L4 Cloud Run; verified e2e 2026-06-23) · `TRACKNET_WSL_SETUP.md` · `README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data_pipeline/`; the field-ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)*
90:2. **? RESOLVED (2026-06-21) ? `NEXT_STEPS.md` refreshed** with a current 2026-06-21 state + a **cross-track prioritized pending list**; the 2026-06-13 owned-model/quality body is kept (clearly marked) as that track's detail.

29. assistant (2026-07-03T16:41:19.867Z)

Now let me look specifically at whether there are two competing sources of truth - NEXT_STEPS P0 item vs the new tracker doc:

30. user (2026-07-03T16:41:20.824Z)

0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only** (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service** (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record: [[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus review:** `docs/ALPHA_LAUNCH_READINESS.md` §7.
0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-included app ? `docs/PACKAGING_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip install` ? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** **(1)** P3 **GPU validation + screencast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB 0-rally tuned-config blocker), **(2)** P2b-install (uv/script + truly-local default config), **(3)** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-

session-packaging-app]]`.
1. ? **`COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged (default-OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, $0). All §8 gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a **$100/?10k cap** (D17).
2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus training feed on it.

31. assistant (2026-07-03T16:41:23.520Z)

Now let me check the §7 of the new ALPHA_LAUNCH_READINESS.md to understand what the source of truth claim is:

32. user (2026-07-03T16:41:24.211Z)

91 ## 7. Deliverables & progress tracker
92
93 Legend: TODO = not started, IN PROGRESS = actively being worked, DONE = shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the row explicitly names a cross-repo batch.
94
95 | ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |
96 |----|-------------|--------------|------------|--------|--------|----|
97 | A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-indexer` | - | No | IN PROGRESS | #451 |
98 | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check, edge auth ON, upload -> process -> jobs -> compile -> download verified | `khelsutra` + `badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |
99 | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer` | A0 | No | TODO | - |
100 | A3 | Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-collector` + vault | A2 | Owner upload/storage scope | TODO | - |
101 | A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2 | Reviewer sign-off | TODO | - |
102 | A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6` floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | TODO | - |
103 | A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the n=15 corpus | `badminton-highlight-indexer` | A2, A4 | No | TODO | - |
104 | A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data availability | TODO | - |
105 | A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly promote/reject person fusion, Gate A served-default, and any other R-series default flips | `badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | TODO | - |
106 | A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner approval for M0.4 scope | TODO | - |
107 | A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-indexer` | A4, A5, A8 | Owner decision | GATED | - |
108 | A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred, document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance | GATED | - |
109 | A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion, `train|reeval` job, and delta-F1 surface | `khelsutra` + `badminton-highlight-indexer` + collector | A1, A3 | Owner D13 reopen | GATED | - |
110 | A13 | Alpha launch review: runbook, privacy/consent copy, known limits, rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11 | Owner launch

decision | TODO | - |

8. Concrete next PR sequence

1. **Docs PR:** land this tracker.
2. **Corpus portability PR:** rebase/portable manifest tooling plus missing-file smoke test.
3. **Hosted alpha PR/batch:** health/auth/upload/process/job/compile/download verification and runbook updates.
4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor refresh.
5. **Ablation/fusion PRs:** fix ablation import, complete 15-video features, rerun promotion/rejection decisions.
6. **Launch decision PR/docs update:** record ship-gate target, C3 status, alpha copy limits, and launch checklist.

9. Open questions

1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or should P10/P11 be reopened before any tester wave?
2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU, GCE L4 VM, or another approved GPU host?
3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains unresolved?
4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a checked-in redacted manifest, or both?
5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every clip, or should the eval loaders support mixed schemas with explicit capability tags?

10. Definition of done

- [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without SSH.
- [ ] Alpha service is gated behind the chosen auth posture and has a documented disable/rollback path.
- [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely on stale local absolute paths.
- [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
- [ ] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
- [ ] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected on the 15-video corpus.
- [ ] Ship-gate target and allowed alpha claims are recorded.
- [ ] WASB C3 status is recorded and reflected in alpha messaging.
- [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for alpha-plus.
- [ ] Owner records GO/NO-GO for the first alpha wave.

33. assistant (2026-07-03T16:41:51.200Z)

Perfect. Now I have all the information I need to do the altitude analysis. Let me compile my findings based on the conventions in DOC_STATUS.md.

Based on my review of the repo's documentation conventions, here are the altitude-related findings for PR #451 (ALPHA_LAUNCH_READINESS.md):

Finding 1: Two competing sources of truth for the same P0 work (NEXT_STEPS.md vs new tracker doc)
- **File:** docs/NEXT_STEPS.md, line 11
- **Summary:** NEXT_STEPS.md P0 item (line 11: "MAKE IT UI-DRIVEN + ALPHA-SHAREABLE") claims to be the "Full record" for UI-driven cloud serving work, while ALPHA_LAUNCH_READINESS.md §7 now claims to be "the source of truth for the concrete next PR sequence." DOC_STATUS.md §2.1 requires updating "ALL inbound references" when a doc's lifecycle changes, but NEXT_STEPS.md P0 remains as a live, parallel reference without demotion or subordination language.
- **Failure scenario:** A contributor updating the P0 item in NEXT_STEPS.md will not auto-sync with the tracker in ALPHA_LAUNCH_READINESS.md §7, causing split information and stale decision

records.

Finding 2: ADR/decision records locked in a DRAFT tracker doc (not in formal ADR log)
- **File:** docs/ALPHA_LAUNCH_READINESS.md, lines 27-35 (§2 "Decisions locked")
- **Summary:** The new doc locks five decisions (D1?D5) inside a DRAFT tracker document, bypassing the ADR mechanism documented in DOC_STATUS.md §3f and exemplified in DECISION_SNAPSHOT_REGRESSION.md §2. Per DOC_STATUS.md §2.1 archival checklist, "decisions locked" should flow through the ADR log (`docs/archives/decisions/DECISIONS.md`). The existing convention shows decisions like "D7?D17" formalized as ADR-014 ratified (2026-06-21, per DOC_STATUS.md §6); D1?D5 here are tactical alpha decisions but are embedded in the DRAFT doc itself.
- **Failure scenario:** If the alpha tracker is deferred or rejected, these decisions (especially D1 "Product alpha and owned-model launch are separate decisions") will not appear in the decision lineage, making future model-launch decisions unable to reference or audit the prior reasoning.

Finding 3: Cross-repo tracker altitude unclear (sibling repos have separate NEXT_STEPS)
- **File:** docs/ALPHA_LAUNCH_READINESS.md, lines 5-6 and §7 cross-repo items (A1, A3, A12)
- **Summary:** DOC_STATUS.md does not specify whether cross-repo trackers should live in one repo's docs/ or in a shared location. The sibling `khelsutra` repo has its own NEXT_STEPS.md and tracked projects (e.g., `STUDIO_MEDIA_LIFECYCLE.md`). ALPHA_LAUNCH_READINESS.md in `badminton-highlight-indexer` now tracks work in `khelsutra`, `sports-data-collector`, and cross-repo deliverables (A1, A3, A12). The khelsutra NEXT_STEPS.md (read above) has no back-reference to this tracker.
- **Failure scenario:** The khelsutra team updating `STUDIO_MEDIA_LIFECYCLE.md` P10/P11 (corpus flywheel; mentioned in ALPHA_LAUNCH_READINESS.md A12) will not see the alpha readiness dependency, and the two repos will track the same work asynchronously.

Finding 4: "Source of truth" claim conflicts with NEXT_STEPS.md language (not demoted)
- **File:** docs/ALPHA_LAUNCH_READINESS.md, line 5 and §7 header (line 91-93)
- **Summary:** The doc's header (line 5) states "Section 7 is the progress tracker and source of truth," but NEXT_STEPS.md P0 (line 11, updated in this PR) still says "Full record: [[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus review:** `docs/ALPHA_LAUNCH_READINESS.md` §7" ? it acknowledges the new tracker exists but doesn't demote its own P0 claim. Per DOC_STATUS.md §2, inbound references should be "repointed" and hierarchy clarified; instead, both docs now claim "source of truth" for overlapping work.
- **Failure scenario:** A PM updating NEXT_STEPS.md P0 (the historically-established entry point) will not realize it is subordinate to the tracker, causing updates to miss the detailed A1?A13 rows.

Finding 5: Doc-status register entry lacks clarity on source-of-truth subordination
- **File:** docs/DOC_STATUS.md, line 33 (the new row for ALPHA_LAUNCH_READINESS.md)
- **Summary:** The added row says this is a "cross-repo alpha readiness tracker created 2026-07-03" and "separates hosted product-alpha work from owned-model/commercial rally-detection gates," but the "Next action" column says "Execute §7 in order" without stating that NEXT_STEPS.md P0 should now be demoted to a summary/reference, not a live tracker. DOC_STATUS.md itself (§2.1) states "flip its row here" when something moves lifecycle, and §2.3 requires "Update ALL inbound references" ? but this PR updates the DOC_STATUS row and adds the README.md entry without demoting NEXT_STEPS.md P0.
- **Failure scenario:** A reviewer reading DOC_STATUS.md will not understand which doc owns the P0 UI-driven work going forward, and the register becomes inconsistent with its own §2 archival checklist.

Findings 1, 4, 5 address the core issue from (3) in the review brief: the doc is "the source of truth" while NEXT_STEPS.md P0 remains live, creating two competing sources of truth for the same P0 work ? and DOC_STATUS.md conventions are not followed to resolve this (demotion, cross-link clarity, inbound-reference repointing).

Finding 3 is the cross-repo altitude issue: an explicit repo-coordination convention is missing from DOC_STATUS.md for where multi-repo work is tracked.

Finding 2 is the decision-recording issue: major decisions (D1?D5) bypass the ADR log, which per the repo's own convention should be the single lineage for locked decisions.

```json
```

```
[
  {
    "file": "docs/NEXT_STEPS.md",
    "line": 11,
    "summary": "NEXT_STEPS.md P0 and ALPHA_LAUNCH_READINESS.md §7 both claim source-of-truth status for the same UI-driven alpha work; DOC_STATUS §2.1 requires demoting inbound references when a doc's lifecycle changes.",
    "failure_scenario": "A contributor updates NEXT_STEPS P0 without seeing ALPHA_LAUNCH_READINESS §7 is now the tracker, causing split information and stale PRs."
  },
  {
    "file": "docs/ALPHA_LAUNCH_READINESS.md",
    "line": 27,
    "summary": "Five decisions (D1?D5) are locked inside this DRAFT tracker doc, bypassing the ADR log (docs/archives/decisions/DECISIONS.md). Per DOC_STATUS §2.1 and DECISION_SNAPSHOT_REGRESSION.md §2 precedent, major decisions should flow through the formal ADR mechanism.",
    "failure_scenario": "If the alpha tracker is deferred/rejected, these decisions (esp. D1 'Product alpha and owned-model launch separate') will not appear in the decision lineage, breaking future model-launch audits."
  },
  {
    "file": "docs/ALPHA_LAUNCH_READINESS.md",
    "line": 5,
    "summary": "Doc claims §7 is 'source of truth' while sibling repos (khelsutra, sports-data-collector) have separate NEXT_STEPS.md and tracked docs. DOC_STATUS.md provides no convention for where cross-repo trackers should live or how to sync sibling NEXT_STEPS files.",
    "failure_scenario": "The khelsutra team updating STUDIO_MEDIA_LIFECYCLE.md P10/P11 (corpus flywheel, dep. A12) won't see the alpha readiness dependency, causing asynchronous tracking across repos."
  },
  {
    "file": "docs/DOC_STATUS.md",
    "line": 33,
    "summary": "The new ALPHA_LAUNCH_READINESS row does not state that NEXT_STEPS.md P0 should now be demoted to a reference. Per §2.1, inbound references must be 'updated' and lifecycle clarified, but the row lacks this subordination language.",
    "failure_scenario": "A reviewer reading DOC_STATUS.md cannot determine which doc owns the P0 work going forward; the register violates its own §2.1 'update ALL inbound references' rule."
  },
  {
    "file": "docs/ALPHA_LAUNCH_READINESS.md",
    "line": 5,
    "summary": "The doc is indexed in README.md and DOC_STATUS.md as a 'cross-repo' tracker, but the DOC_STATUS §3a register entry (line 33) treats it as a single-repo project. The 'Repo / owner' column in §7 lists multiple repos (A1, A3, A12), conflicting with the register classification.",
    "failure_scenario": "A contributor looking at DOC_STATUS §3a (which classifies by single-repo project ownership) will misunderstand ALPHA_LAUNCH_READINESS as owned by badminton-highlight-indexer alone, missing khelsutra and collector dependencies."
  }
]
```

```
```